



模式识别与计算机视觉

Chapter 9: Interest Points: Descriptor

王利民

媒体计算课题组

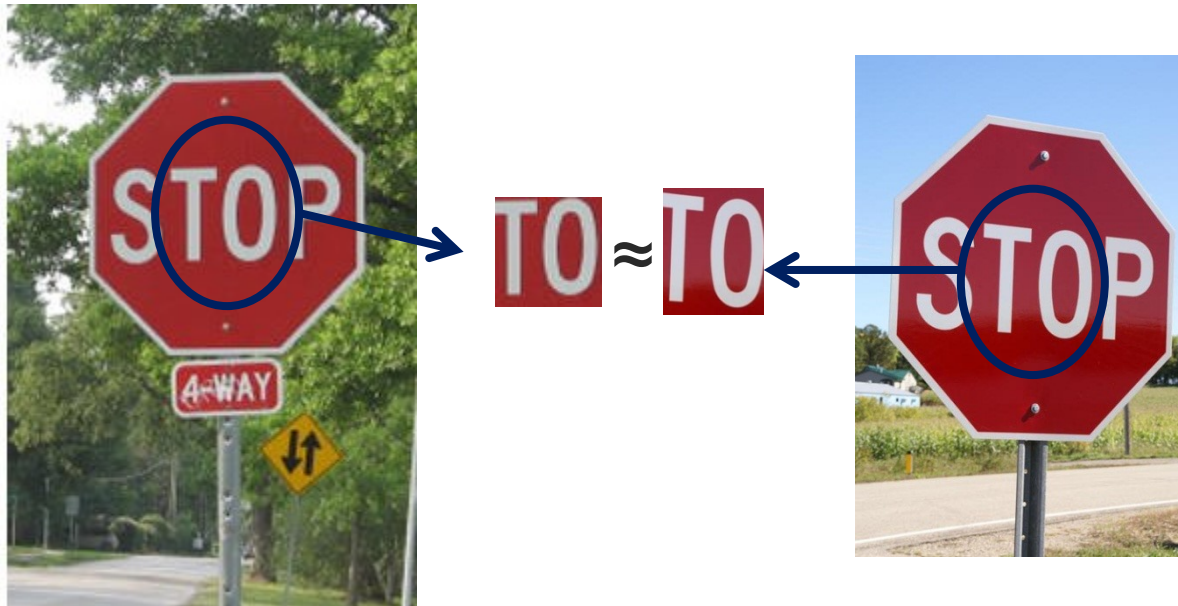
<http://mcg.nju.edu.cn/>



Correspondence and alignment

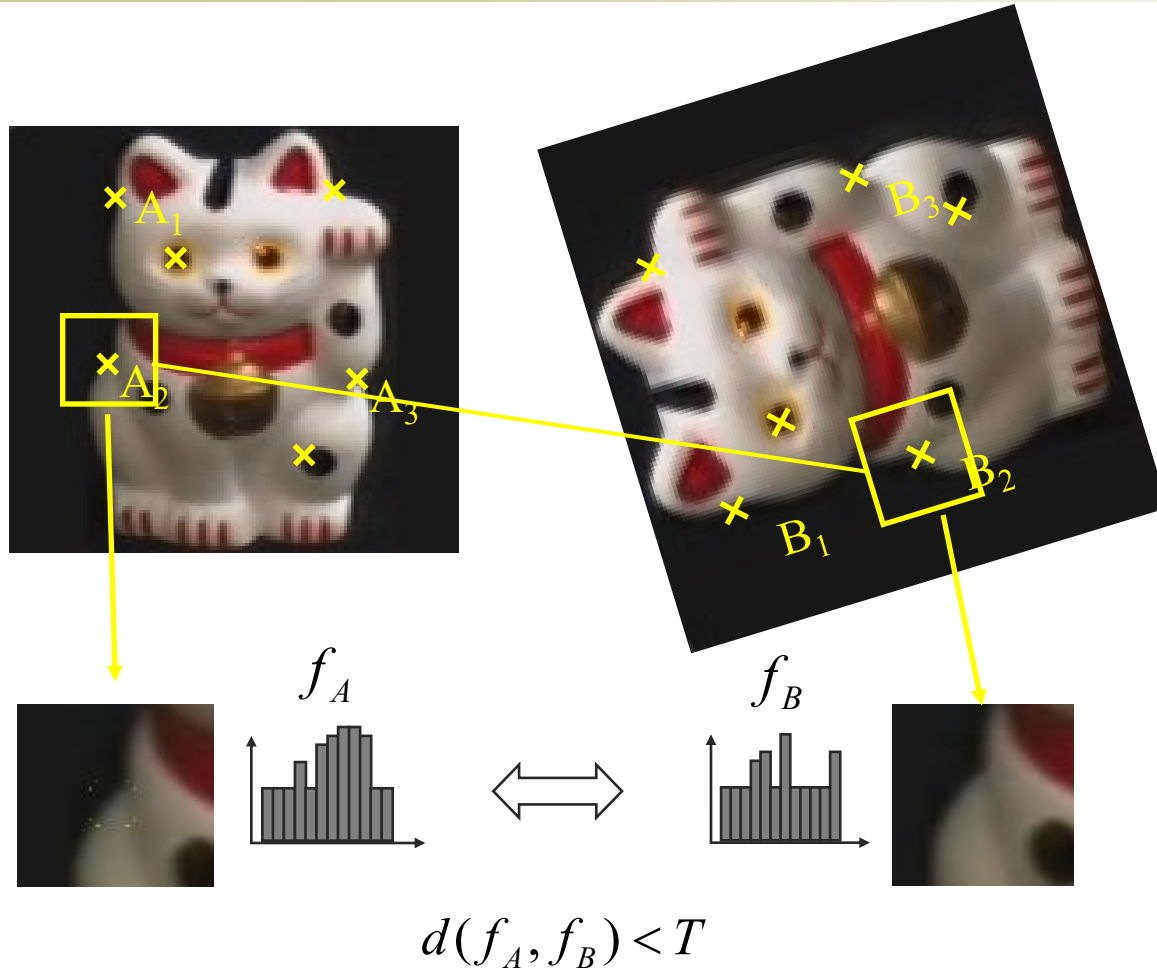


Correspondence: matching points, patches, edges, or regions across images





Recap: Keypoint Matching



1. Find a set of distinctive keypoints
2. Define a region around each keypoint
3. Extract and normalize the region content
4. Compute a local descriptor from the normalized region
5. Match local descriptors



Today's class



- Normalization
 - Orientation normalization
 - Affine invariant feature extraction
- Local descriptor
 - SIFT, SURF, GIST
- Binary descriptor
 - LBP, BRIEF
- CNN based descriptor
 - MatchNet, DeepCompare, DeepDesc, LIFT

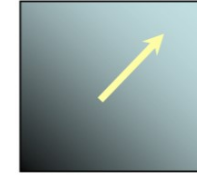
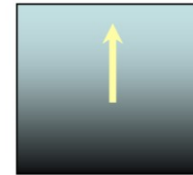


Rotation Invariant Descriptors



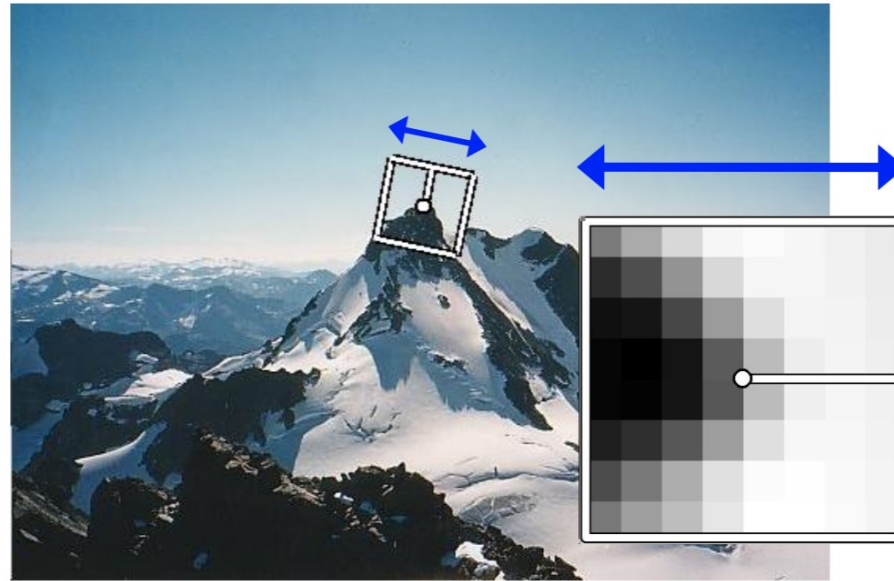
- Find local orientation

- Dominant direction of gradient for the image patch



- Rotate patch according to this angle

- This puts the patches into a canonical orientation.



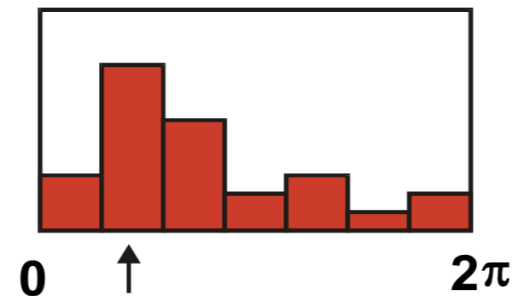
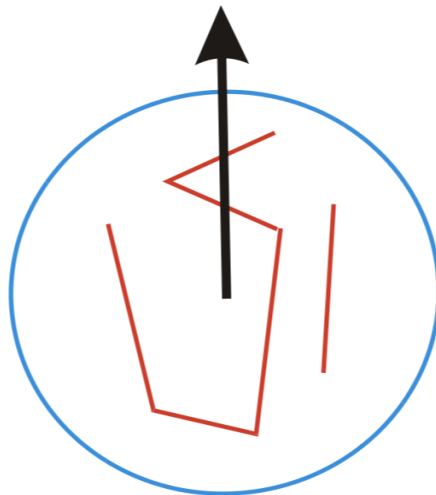


Orientation Normalization



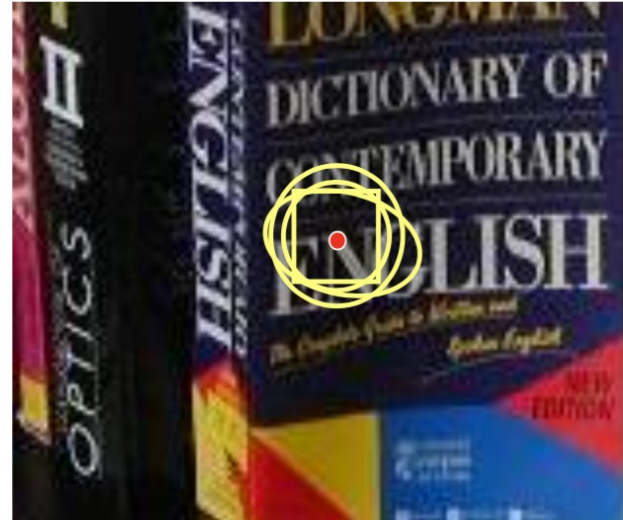
- Compute orientation histogram
- Select dominant orientation
- Normalize: rotate to fixed orientation

[Lowe, SIFT, 1999]



Slide adapted from David Lowe

The Need for Invariance



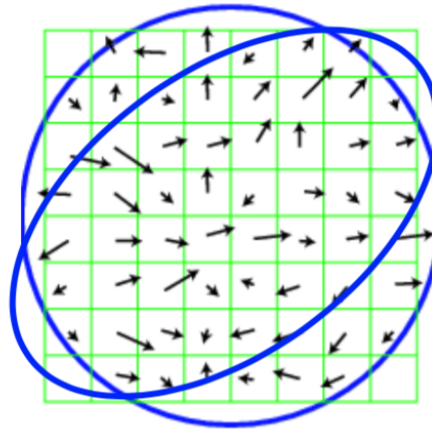
- Up to now, we had invariance to
 - Translation
 - Scale
 - Rotation
- Not sufficient to match regions under viewpoint changes
 - For this, we need also affine adaptation



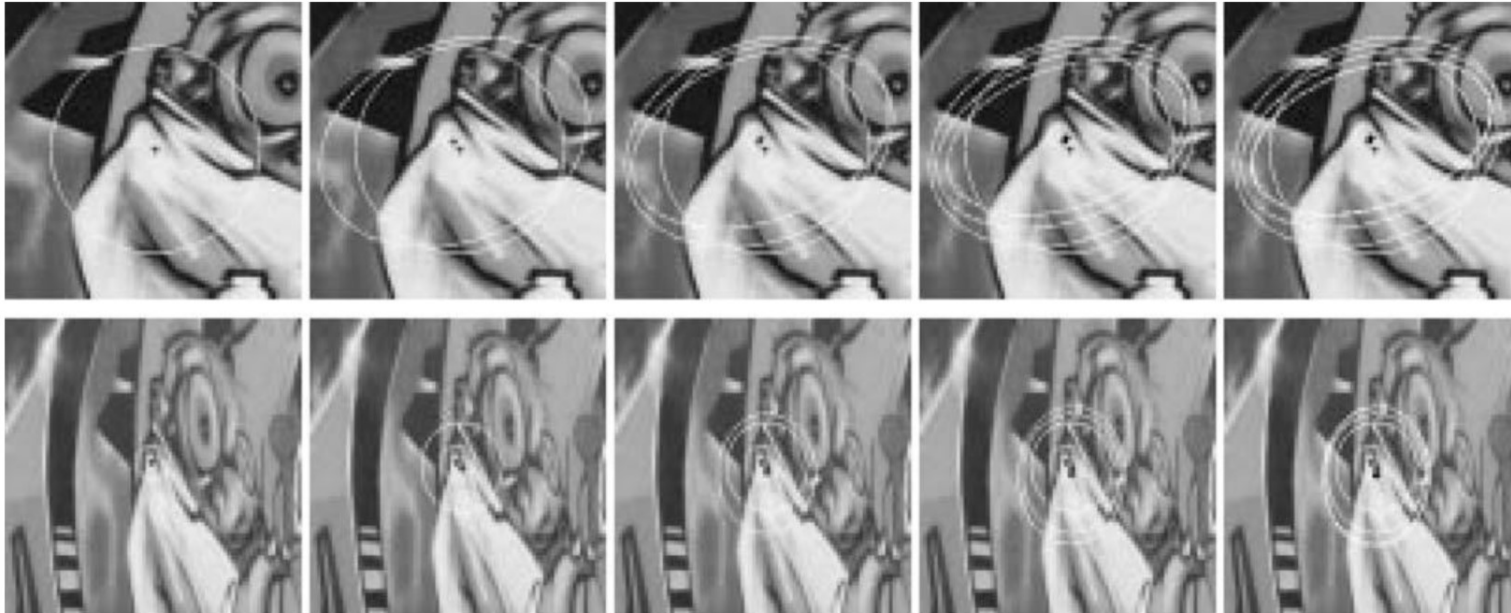
Affine Adaption



- **Problem:**
 - Determine the characteristic shape of the region.
 - Assumption: shape can be described by “local affine frame”.
- **Solution: iterative approach**
 - Use a circular window to compute second moment matrix.
 - Compute eigenvectors to adapt the circle to an ellipse.
 - Recompute second moment matrix using new window and iterate...



Iterative Adaption



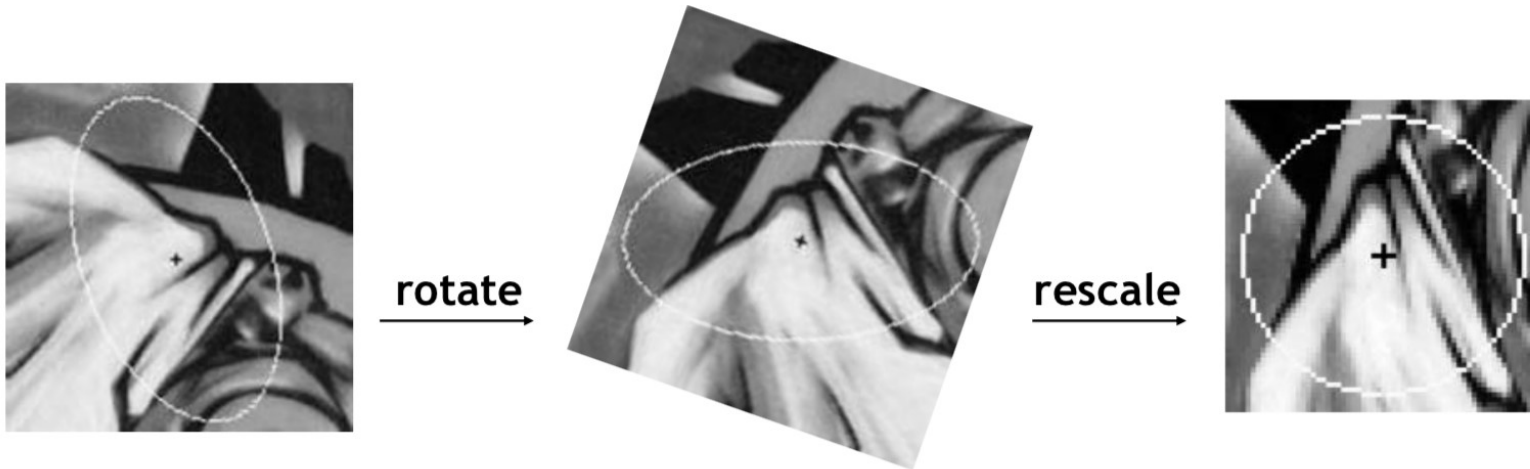
1. Detect keypoints, e.g. multi-scale Harris
2. Automatically select the scales
3. Adapt affine shape based on second order moment matrix
4. Refine point location

K. Mikolajczyk and C. Schmid, [Scale and affine invariant interest point detectors](#),
IJCV 60(1):63-86, 2004.

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Slide credit: Tinne Tuytelaars

Affine Normalization

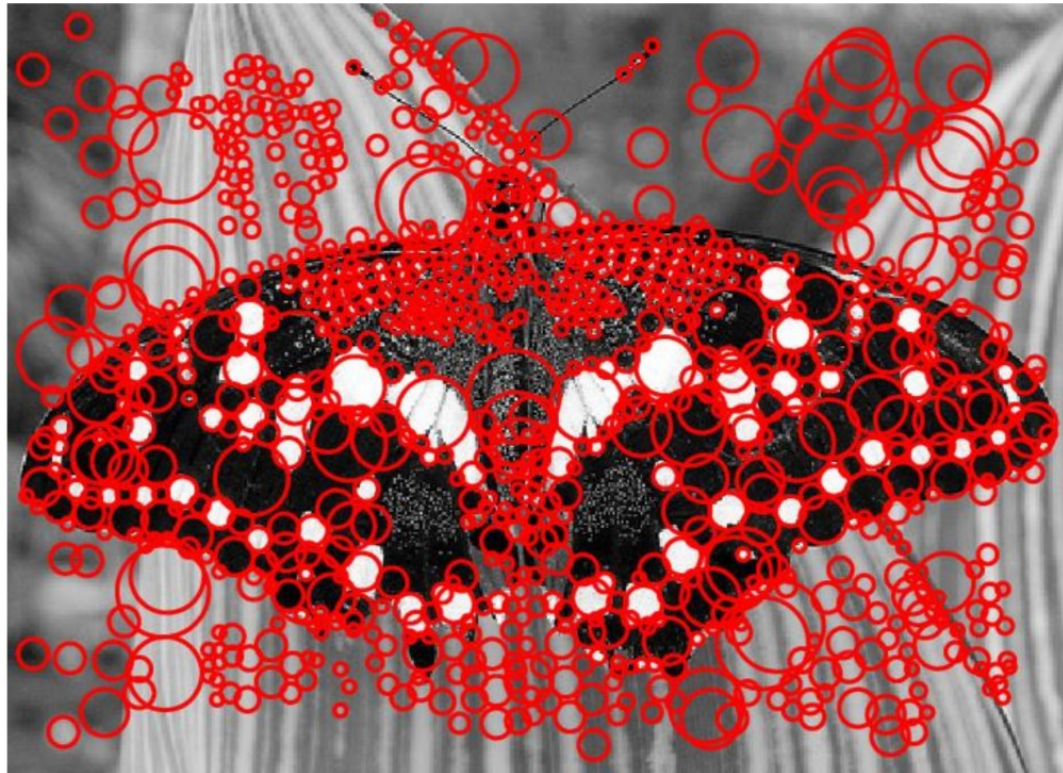


- **Steps**

- Rotate the ellipse's main axis to horizontal
- Scale the x axis, such that it forms a circle



Affine Adaption Example



Scale-invariant regions (blobs)



Affine Adaption Example



Affine-adapted blobs



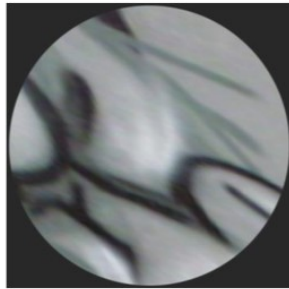
Summary: Affine Invariance



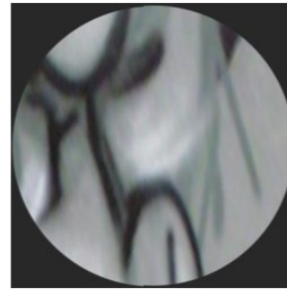
Extract affine regions



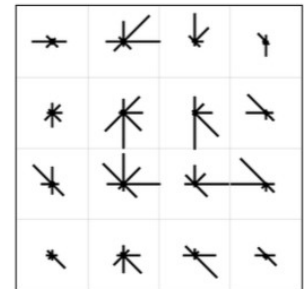
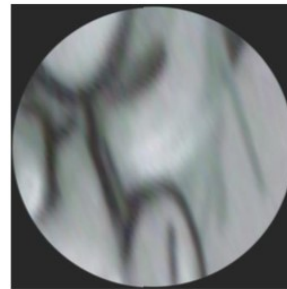
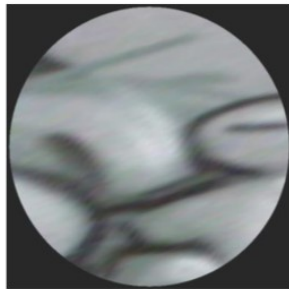
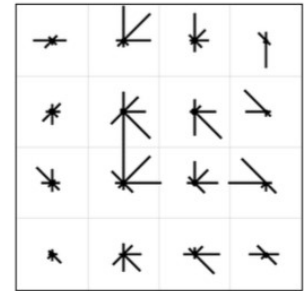
Normalize regions



Eliminate rotational ambiguity



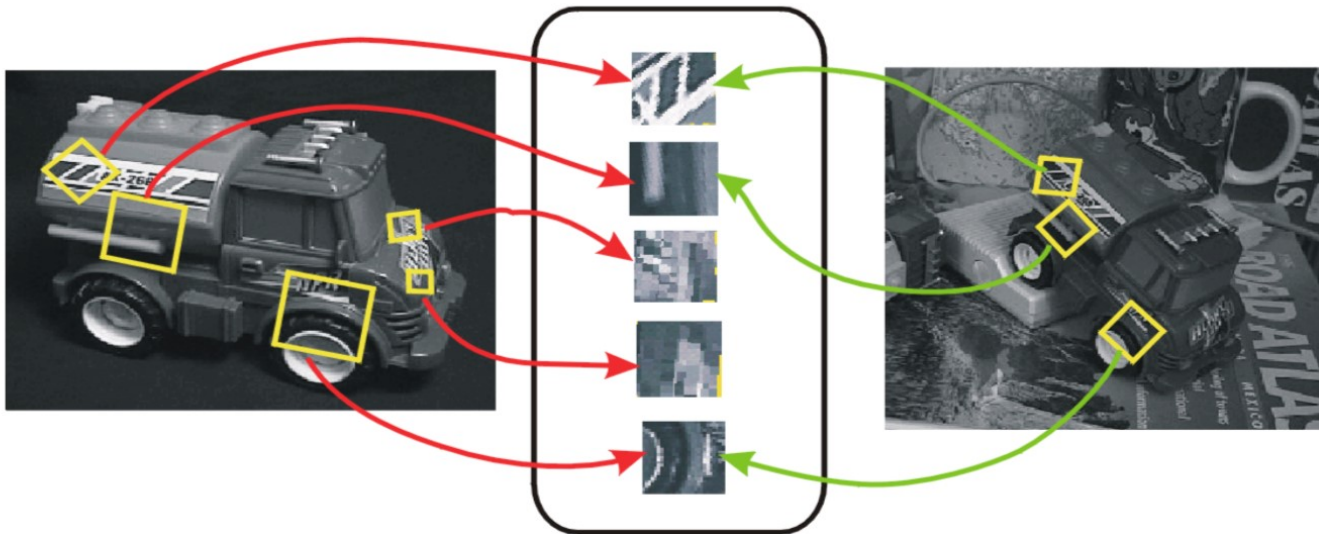
Compare descriptors



Invariance vs. Covariance



- **Invariance:**
 - $\text{features}(\text{transform}(\text{image})) = \text{features}(\text{image})$
- **Covariance:**
 - $\text{features}(\text{transform}(\text{image})) = \text{transform}(\text{features}(\text{image}))$



Covariant detection $\Rightarrow$ invariant description



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 - MatchNet, DeepCompare, DeepDesc, LIFT

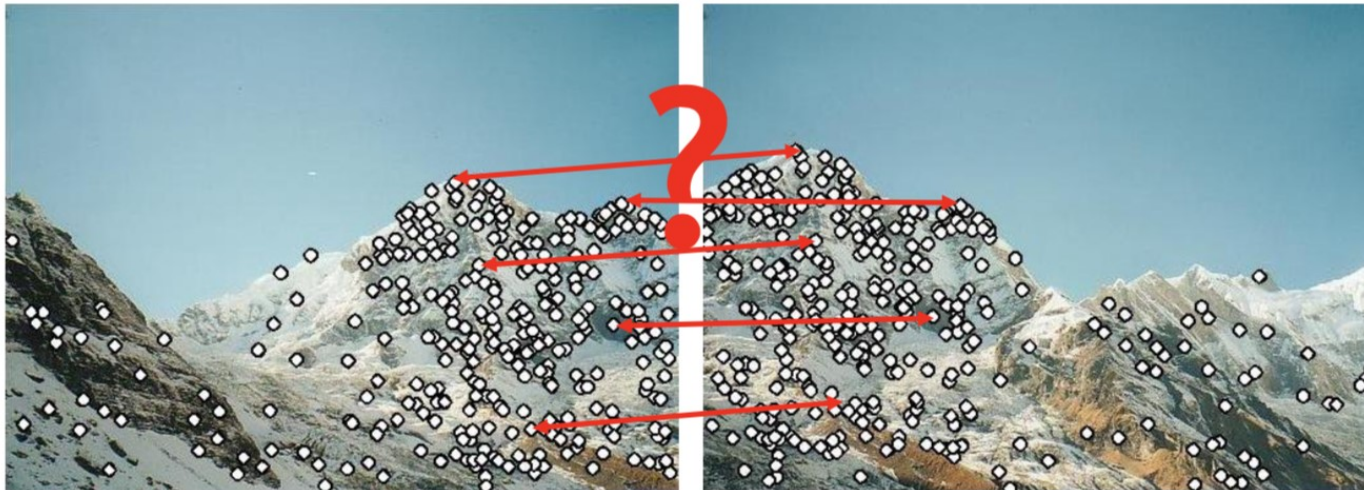


Local Descriptor



- We know how to detect points
- Next question:

How to describe them for matching?

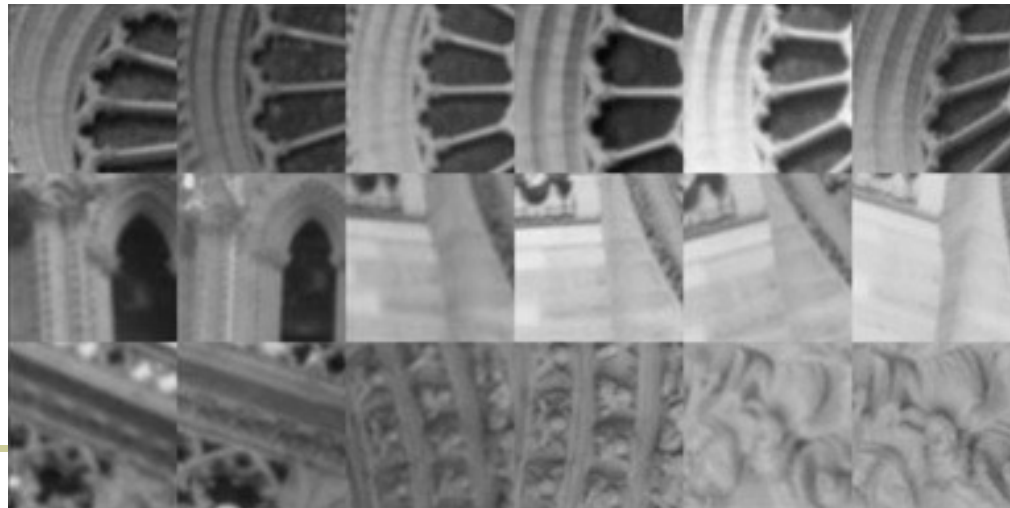


Point descriptor should be:

1. Invariant
2. Distinctive

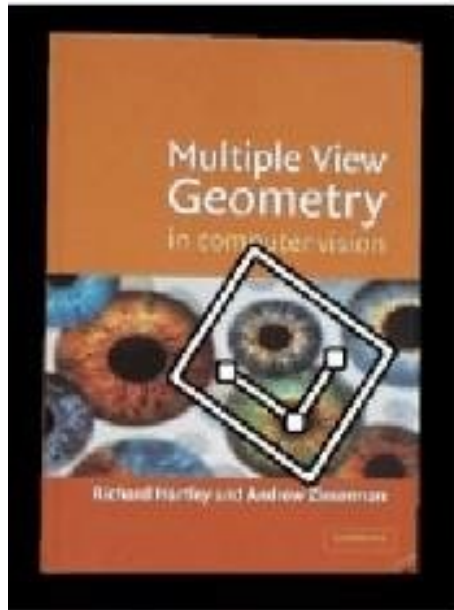


What is the best descriptor for an image feature?



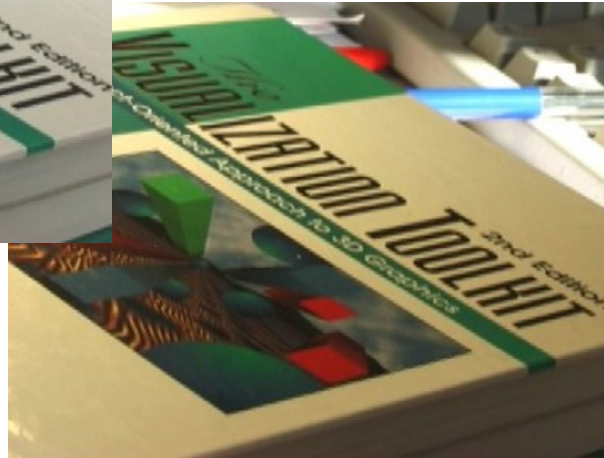
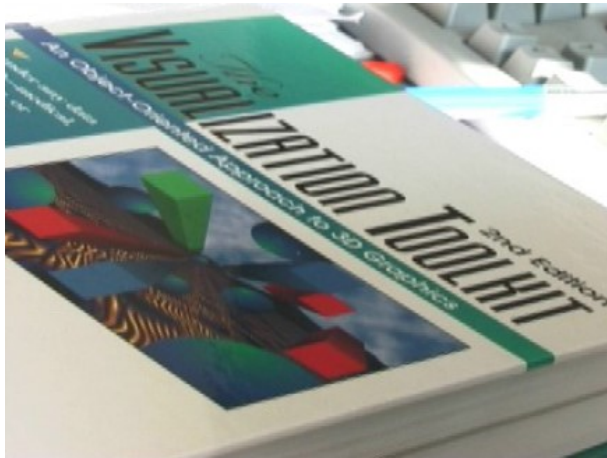


Geometric transformations





Photometric transformations





Local Descriptor

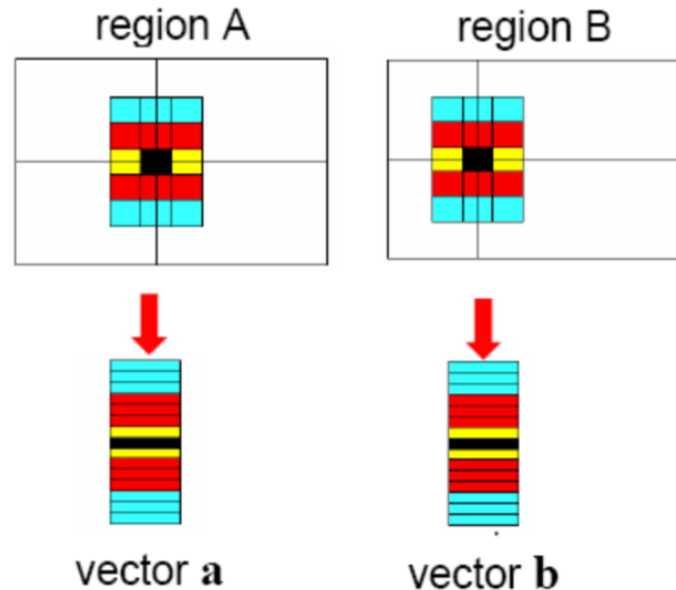
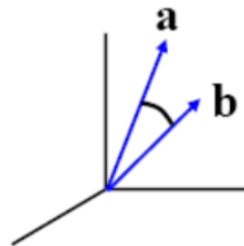


- Simplest descriptor: list of intensities within a patch.
- What is this going to be invariant to?

**Perfectly fine if geometry and appearance is unchanged
(a.k.a. template matching)**

Write regions as vectors

$A \rightarrow \mathbf{a}, B \rightarrow \mathbf{b}$



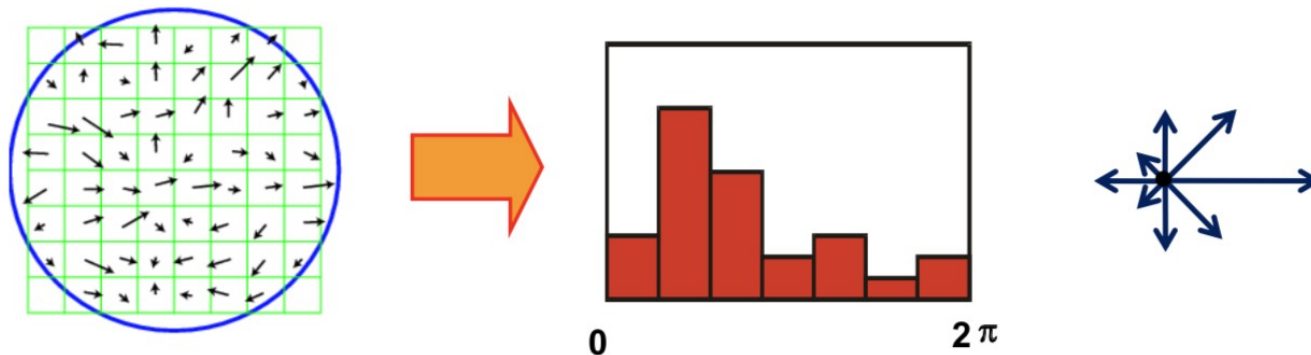
Feature Descriptor



- **Disadvantage of patches as descriptors:**
 - Small shifts can affect matching score a lot



- **Solution: histograms**

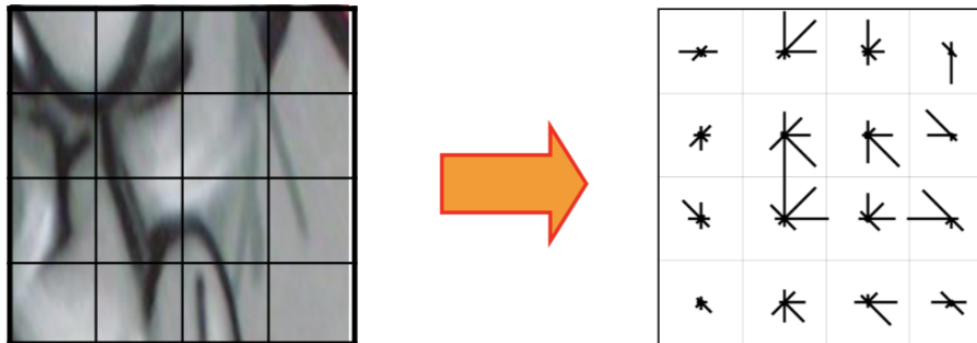




Feature Descriptor: SIFT



- **S**cale **I**nvariant **F**eature **T**ransform
- **D**escriptor computation:
 - Divide patch into 4x4 sub-patches: 16 cells
 - Compute histogram of gradient orientations (8 reference angles) for all pixels inside each sub-patch
 - Resulting descriptor: $4 \times 4 \times 8 = 128$ dimensions



David G. Lowe. ["Distinctive image features from scale-invariant keypoints."](#)
IJCV 60 (2), pp. 91-110, 2004.



SIFT Properties



- **Extraordinarily robust matching technique**
 - Can handle changes in viewpoint up to ~60 deg. out-of-plane rotation
 - Can handle significant changes in illumination
 - Sometimes even day vs. night (below)
 - Fast and efficient—can run in real time
 - Lots of code available
 - http://people.csail.mit.edu/albert/ladypack/wiki/index.php/Known_implementations_of_SIFT



Slide credit: Steve Seitz



Summary: SIFT

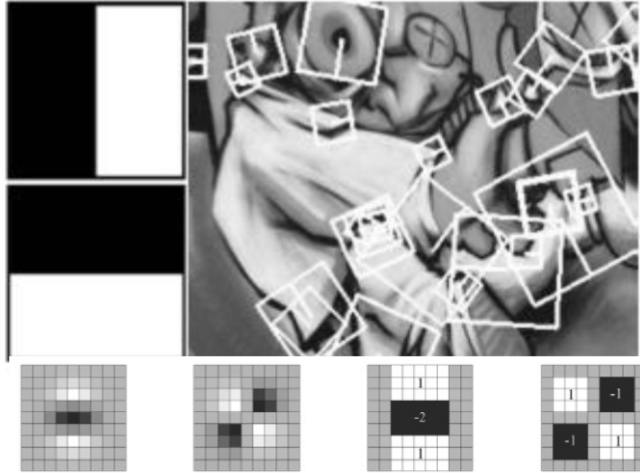


- One image yields:
 - n 2D points giving positions of the patches
 - $[n \times 2 \text{ matrix}]$
 - n scale parameters specifying the size of each patch
 - $[n \times 1 \text{ vector}]$
 - n orientation parameters specifying the angle of the patch
 - $[n \times 1 \text{ vector}]$
 - n 128-dimensional descriptors: each one is a histogram of the gradient orientations within a patch
 - $[n \times 128 \text{ matrix}]$





Feature Descriptor: SURF



- **Fast approximation of SIFT idea**

- Efficient computation by 2D box filters & integral images
⇒ 6 times faster than SIFT
- Equivalent quality for object identification
- <http://www.vision.ee.ethz.ch/~surf>

Well received! **32747**

- More than 8000 citations.
- CVIU Most Cited Paper
- Koenderink Prize of ECCV'16

GPU implementation available

- Feature extraction @ 100Hz
(detector + descriptor, 640×480 img)
- <http://homes.esat.kuleuven.be/~ncorneli/gpusurf/>

Herbert Bay et al., SURF: **S**peeded **U**p **R**obust **F**eature, in ECCV 2006



SURF: Keypoint Detection



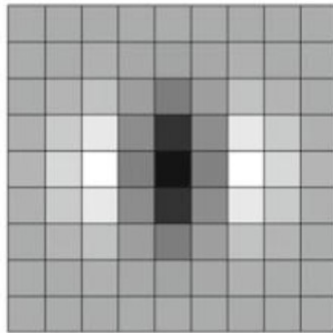
- Uses determinant of Hessian matrix
- Approximate 2nd derivatives in Hessian matrix with box filters

$$H(x, \sigma) = \begin{bmatrix} L_{xx}(x, \sigma) & L_{xy}(x, \sigma) \\ L_{xy}(x, \sigma) & L_{yy}(x, \sigma) \end{bmatrix} \longrightarrow \hat{H}(x, \sigma) = \begin{bmatrix} D_{xx}(x, \sigma) & D_{xy}(x, \sigma) \\ D_{xy}(x, \sigma) & D_{yy}(x, \sigma) \end{bmatrix}$$

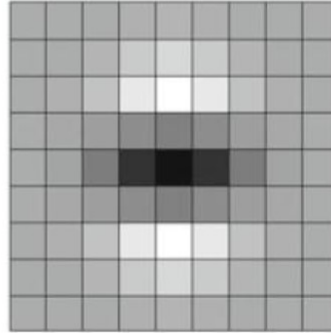
$$R(x, \sigma) = L_{xx}L_{yy} - L_{xy}^2 \approx D_{xx}D_{yy} - (0.9D_{xy}^2)$$



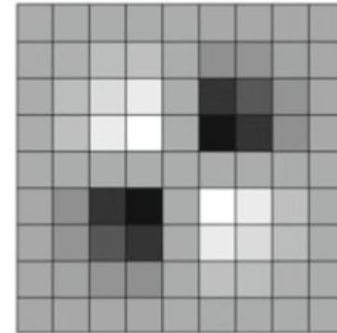
SURF: Keypoint Detection



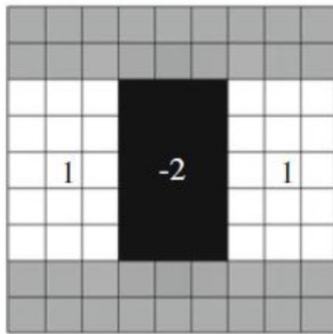
L_{xx}



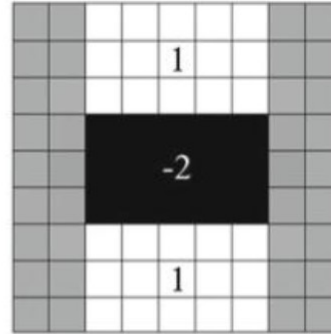
L_{yy}



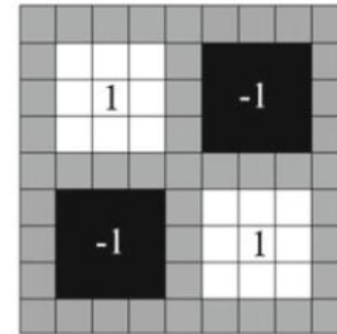
L_{xy}



D_{xx}



D_{yy}



D_{xy}



Integral Image



original
image

$$I(x, y)$$

1	5	2
2	4	1
2	1	1

$$A(x, y)$$

1	6	8
3	12	15
5	15	19

integral
image

$$A(x, y) = \sum_{x' \leq x, y' \leq y} I(x', y')$$



Integral Image



original
image

$I(x, y)$		
1	5	2
2	4	1
2	1	1

$A(x, y)$

1	6	8
3	12	15
5	15	19

integral
image

$$A(x, y) = \sum_{x' \leq x, y' \leq y} I(x', y')$$

Can find the **sum** of any block using **3** operations

$$A(x_1, y_1, x_2, y_2) = A(x_2, y_2) - A(x_1, y_2) - A(x_2, y_1) + A(x_1, y_1)$$



What is the sum of the bottom right 2x2 square?



$$A(x_1, y_1, x_2, y_2) = A(x_2, y_2) - A(x_1, y_2) - A(x_2, y_1) + A(x_1, y_1)$$

$I(x, y)$

1	5	2
2	4	1
2	1	1

image

$A(x, y)$

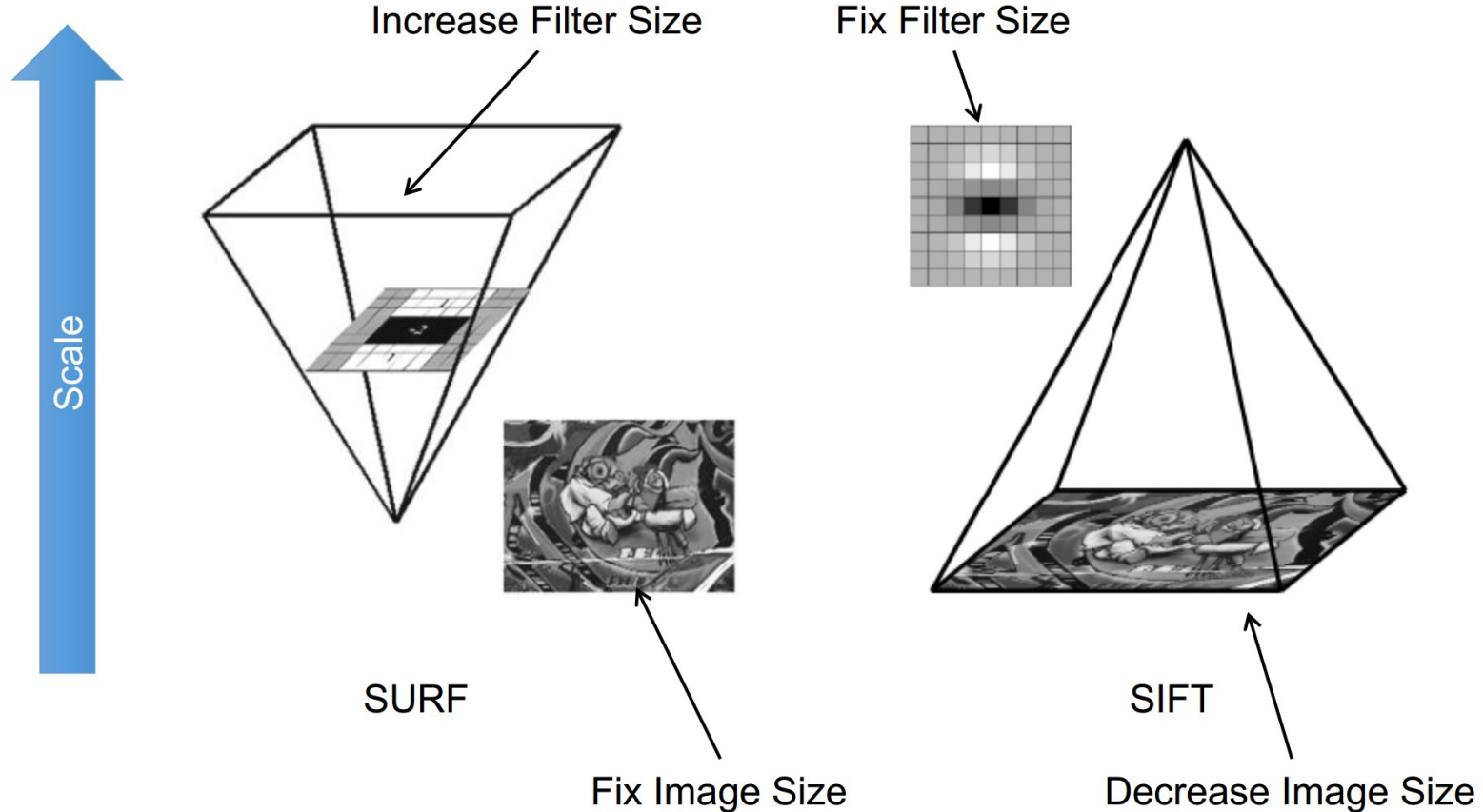
1	6	8
3	12	15
5	15	19

integral image

$$\begin{aligned} A(1, 1, 3, 3) &= A(3, 3) - A(1, 3) - A(3, 1) + A(1, 1) \\ &= 19 - 8 - 5 + 1 \\ &= 7 \end{aligned}$$

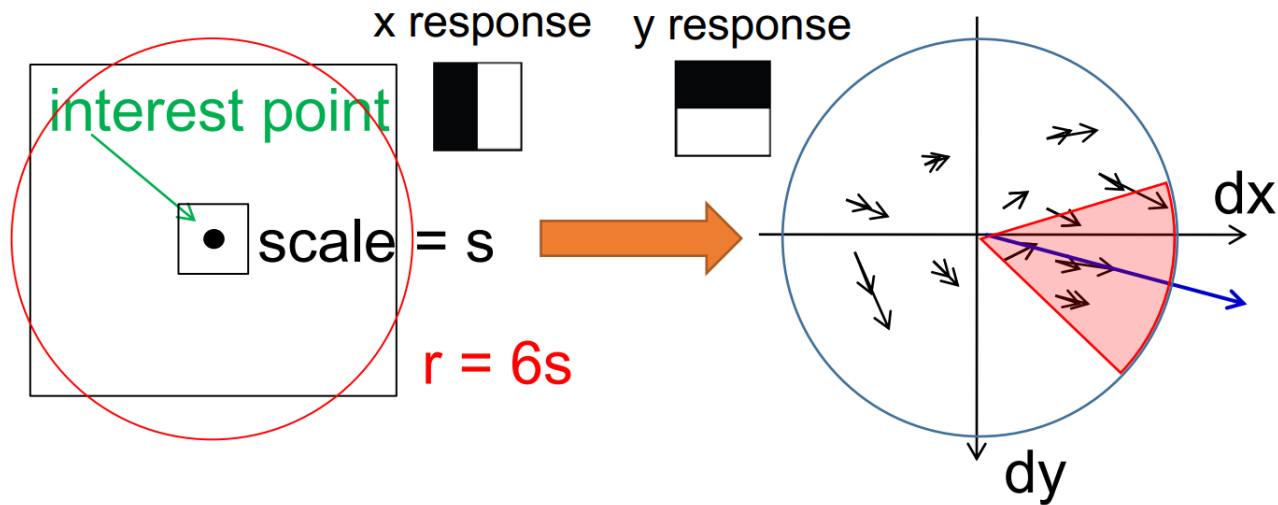


SURF vs. SIFT: Scale Space





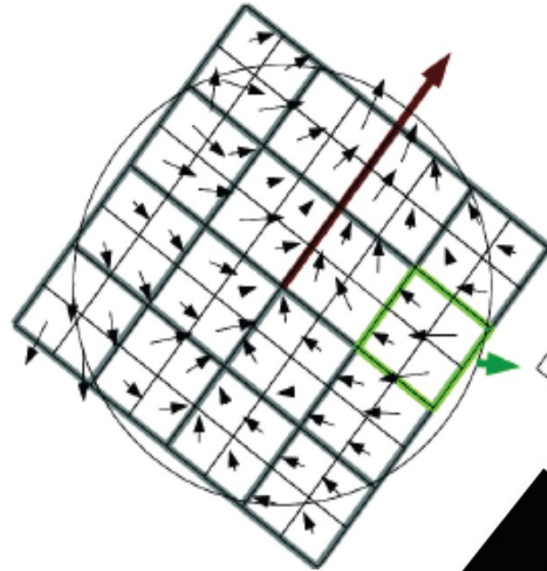
Dominant Orientation Estimation



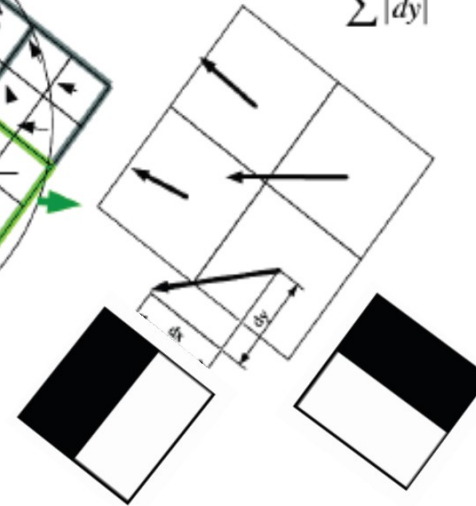
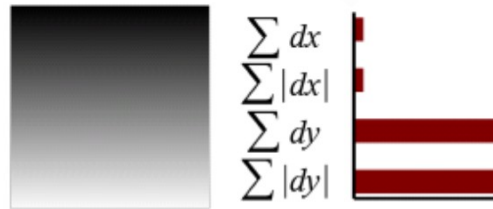
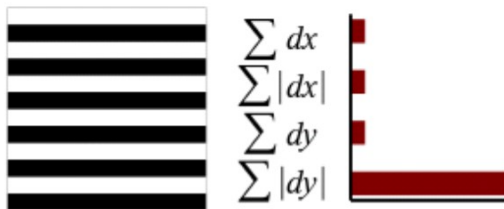
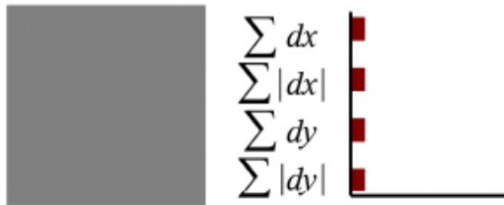
- The Haar wavelet responses (x and y) are represented as vectors.
- Sum all responses within a sliding orientation window covering an angle of 60 degree.
- The longest vector is the dominant orientation



SURF: Descriptor Extraction



$$\begin{matrix} \sum dx \\ \sum |dx| \\ \sum dy \\ \sum |dy| \end{matrix}$$

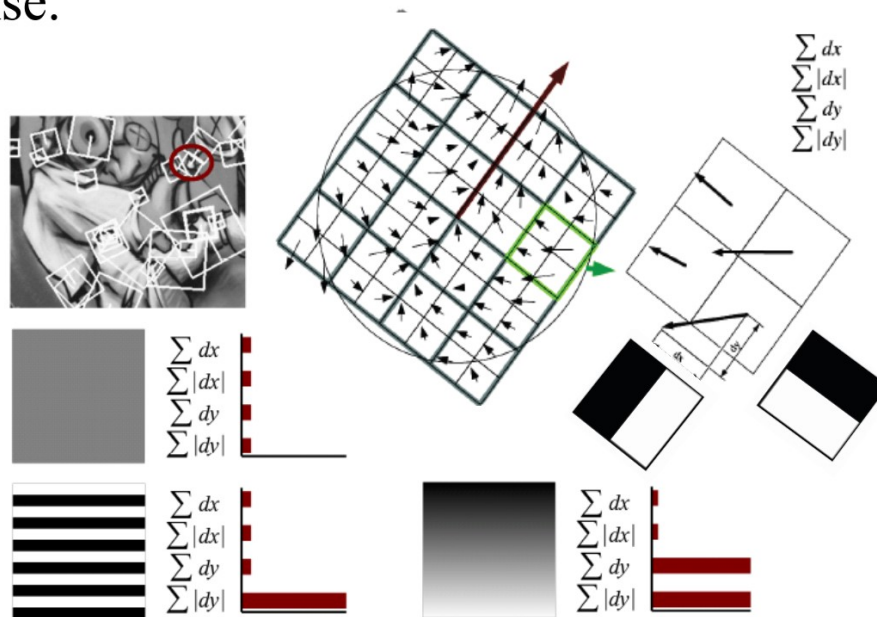




SURF: Descriptor Extraction



1. Split the interest region (20s x 20s) into 4 x 4 square sub-regions.
2. Calculate Haar wavelet responses dx and dy , and weight the responses with a Gaussian kernel.
3. Sum the response over each sub-region for dx and dy , then sum the absolute value of response.
4. Concatenate summation results in all sub-regions, forming a 64D SURF descriptor.





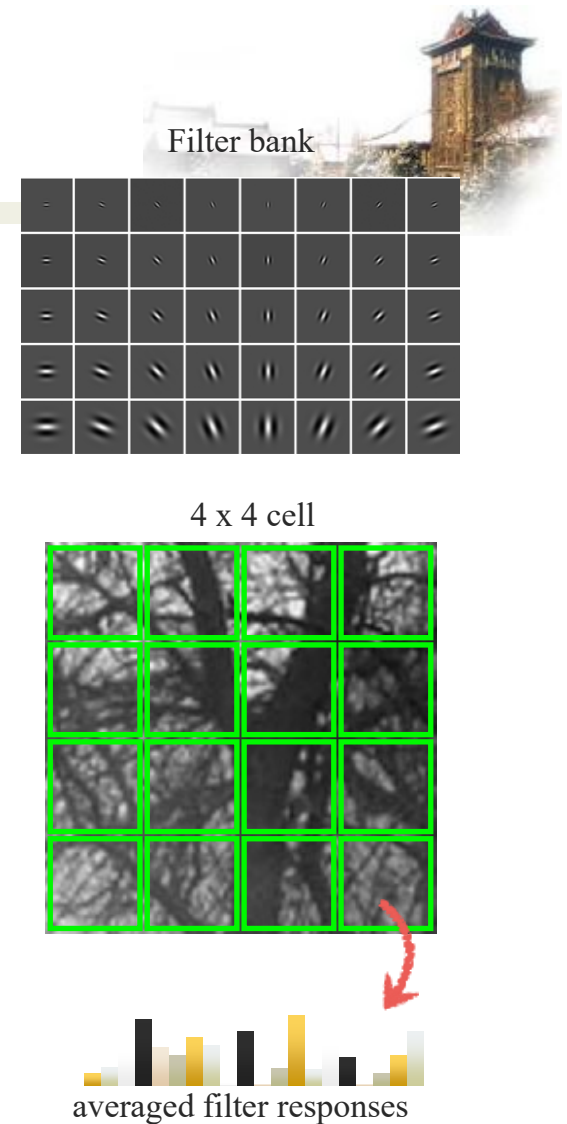
Summary: SURF



- Approximation yet can be computed much faster.
 - relying on **integral images** for image convolutions
 - building on the strengths of the **leading existing detectors and descriptors**
 - **simplifying** these methods to the essential
- Combination of novel detection, description, and matching steps.

GIST

1. Compute filter responses (filter bank of Gabor filters)
2. Divide image patch into 4 x 4 cells
3. Compute filter response averages for each cell
4. Size of descriptor is $4 \times 4 \times N$, where N is the size of the filter bank



Oliva, A., & Torralba, A., Modeling the Shape of the Scene: a Holistic Representation of the Spatial Envelope. IJCV 2001.

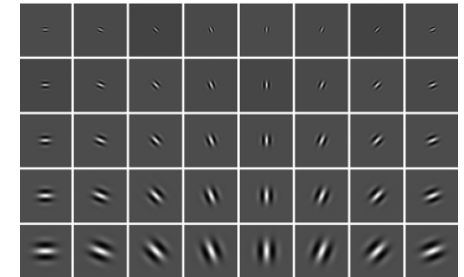


Summary: GIST

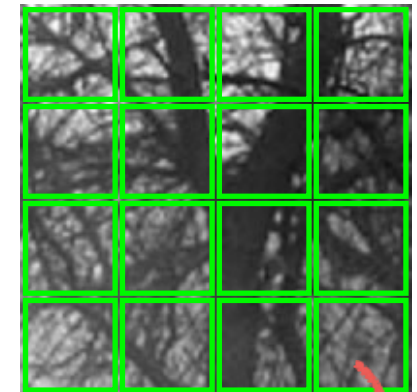


1. Compute filter responses (filter bank of Gabor filters)
2. Divide image patch into 4 x 4 cells
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4. Size of descriptor is 4 x 4 x N, where N is the size of the filter bank

Filter bank



4 x 4 cell



What is the GIST descriptor encoding?

Rough spatial distribution of image gradients





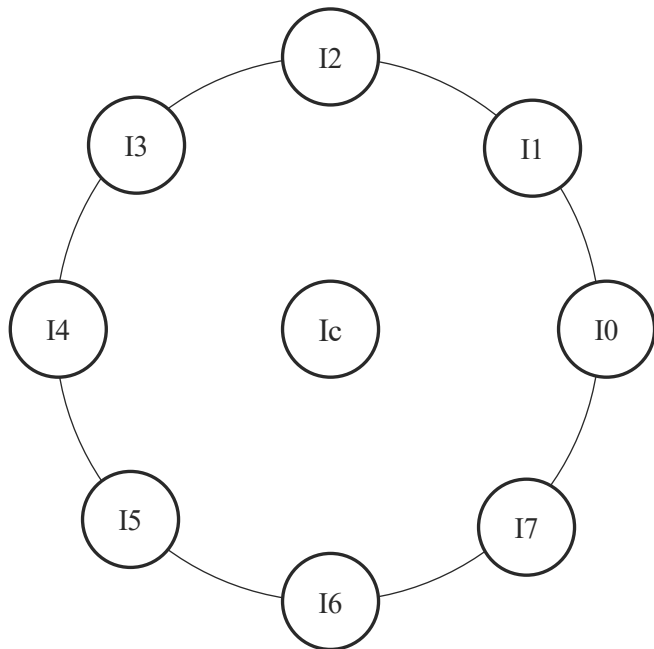
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- **Binary descriptor**
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- CNN Based descriptor
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Center Symmetric Local Binary Patterns



$$\begin{aligned} \text{CS-LBP} = & \text{sign}(I_0 - I_4 - t)2^0 + \\ & \text{sign}(I_1 - I_5 - t)2^1 + \\ & \text{sign}(I_2 - I_6 - t)2^2 + \\ & \text{sign}(I_3 - I_7 - t)2^3 \end{aligned}$$

$$\begin{aligned} \text{LBP} = & \text{sign}(I_0 - I_c)2^0 + \\ & \text{sign}(I_1 - I_c)2^1 + \\ & \text{sign}(I_2 - I_c)2^2 + \\ & \text{sign}(I_3 - I_c)2^3 + \\ & \text{sign}(I_4 - I_c)2^4 + \\ & \text{sign}(I_5 - I_c)2^5 + \\ & \text{sign}(I_6 - I_c)2^6 + \\ & \text{sign}(I_7 - I_c)2^7 \end{aligned}$$

Robustness to illumination.

CSLBP [Heikkilä '09]



Binary Robust Independent Elementary Features



Descriptor construction: intensity test
between given point pairs:

$$f_n(P) = [\tau(P; x_1, y_1), \dots, \tau(P; x_n, y_n)] \in \{0, 1\}^n$$

$$\tau(P; x, y) = \begin{cases} 1, & P(x) > P(y) \\ 0, & P(x) \leq P(y) \end{cases}$$

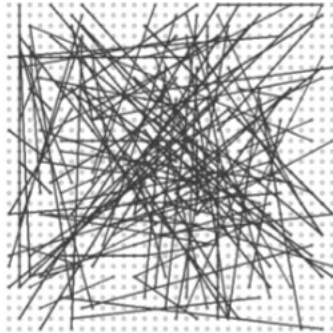
Point pairs: given by random sampling

$$x \square G(0, \frac{S^2}{25}), y \square G(0, \frac{S^2}{25})$$

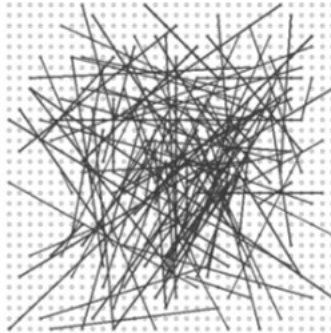
- Simple, fast, moderate performance
- First binary descriptor for patch matching



Binary Robust Independent Elementary Features



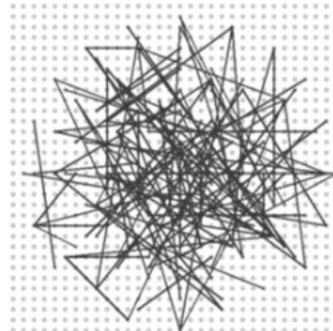
G I



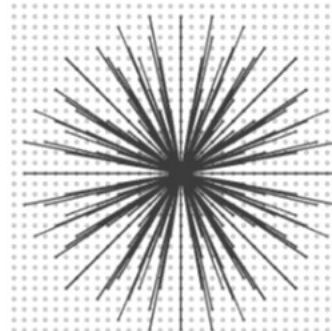
G II



G III



G IV



G V



Binary Robust Independent Elementary Features

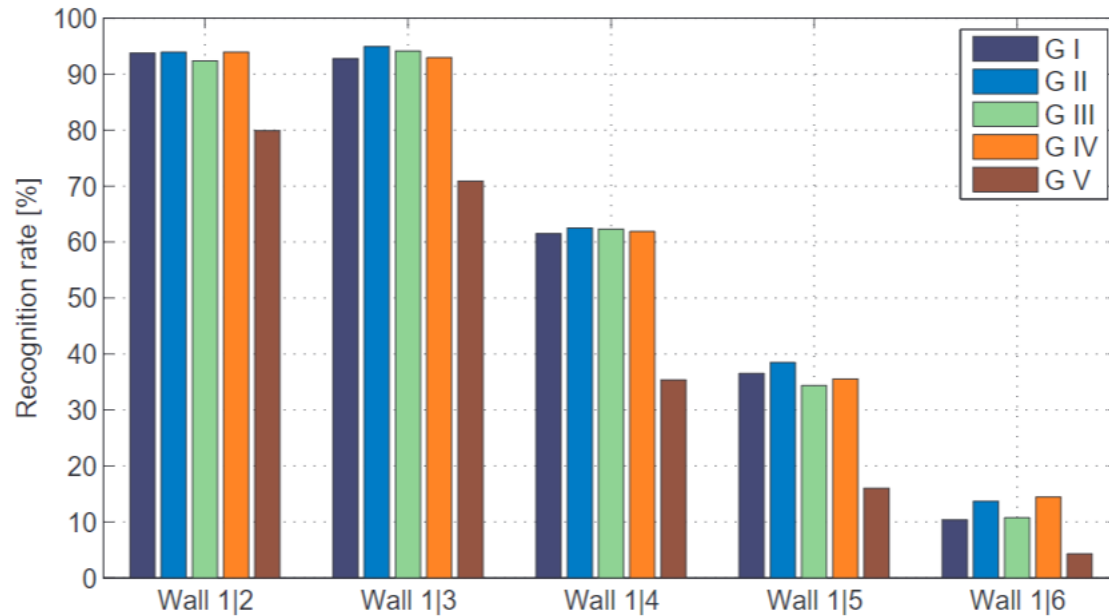


Fig. 3. Recognition rate for the five different test geometries introduced in section 3.2.



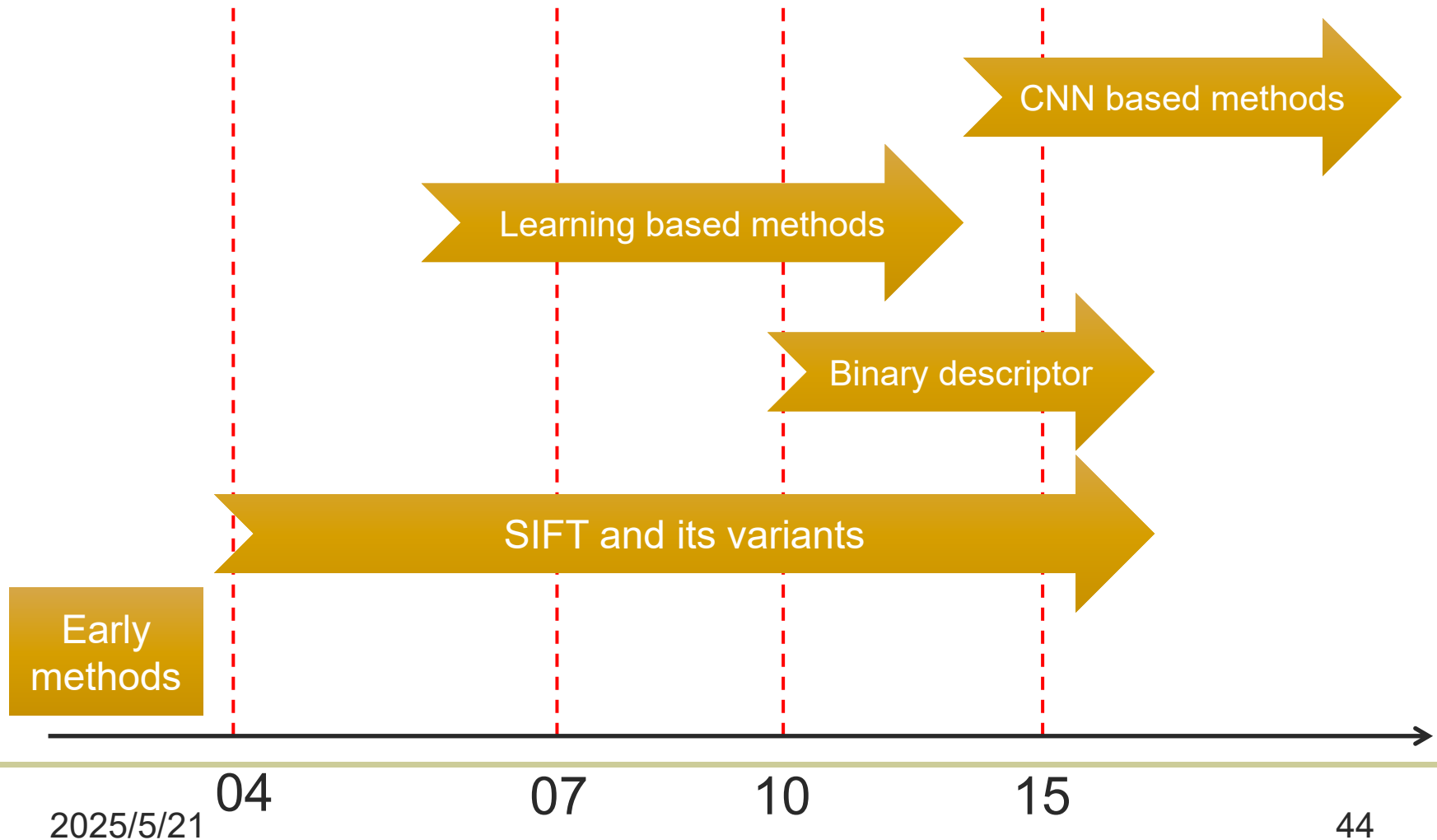
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 - MatchNet, DeepCompare, DeepDesc, LIFT



Local Descriptors: Trend





A Deep Casualty?



Distinctive image features from scale-invariant keypoints

[PDF] from robo.fish

Authors David G Lowe

Publication date 2004/11/1

Journal International journal of computer vision

Volume 60

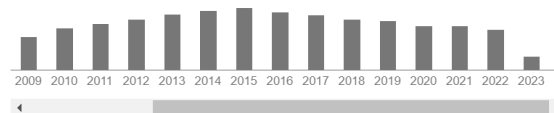
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Description This paper presents a method for extracting distinctive invariant features from images that can be used to perform reliable matching between different views of an object or scene. The features are invariant to image scale and rotation, and are shown to provide robust matching across a substantial range of affine distortion, change in 3D viewpoint, addition of noise, and change in illumination. The features are highly distinctive, in the sense that a single feature can be correctly matched with high probability against a large database of features from many images. This paper also describes an approach to using these features for object recognition. The recognition proceeds by matching individual features to a database of features from known objects using a fast nearest-neighbor algorithm, followed by a Hough transform to identify clusters belonging to a single object, and finally performing verification through ...

Total citations Cited by 70589



Scholar articles [Distinctive image features from scale-invariant keypoints](#)
DG Lowe - International journal of computer vision, 2004
[Cited by 70589](#) [Related articles](#) [All 148 versions](#)

- The SIFT paper is the most cited computer vision paper **ever**.
- But it's not as dominant as it once was.



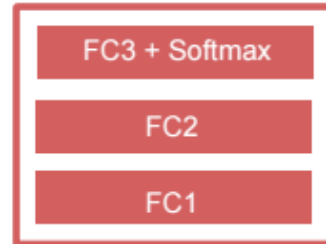
MatchNet



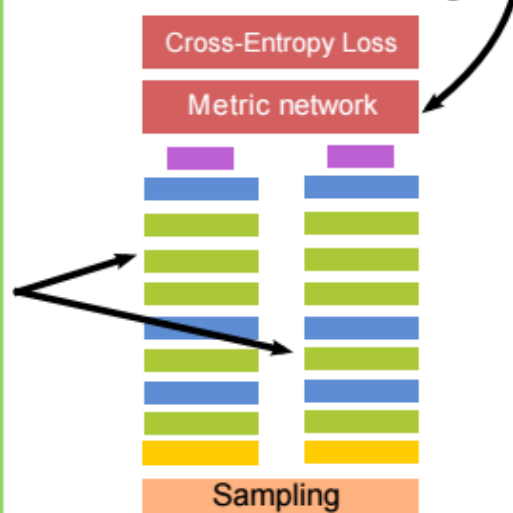
A: Feature network



B: Metric network



C: MatchNet in training



- Simultaneously learn the descriptor and the metric
- Siamese Feature descriptor network
- Metric network on top
- Cross-entropy loss, transfer matching problem to classification problem
- Train time: 1 day – 1 week



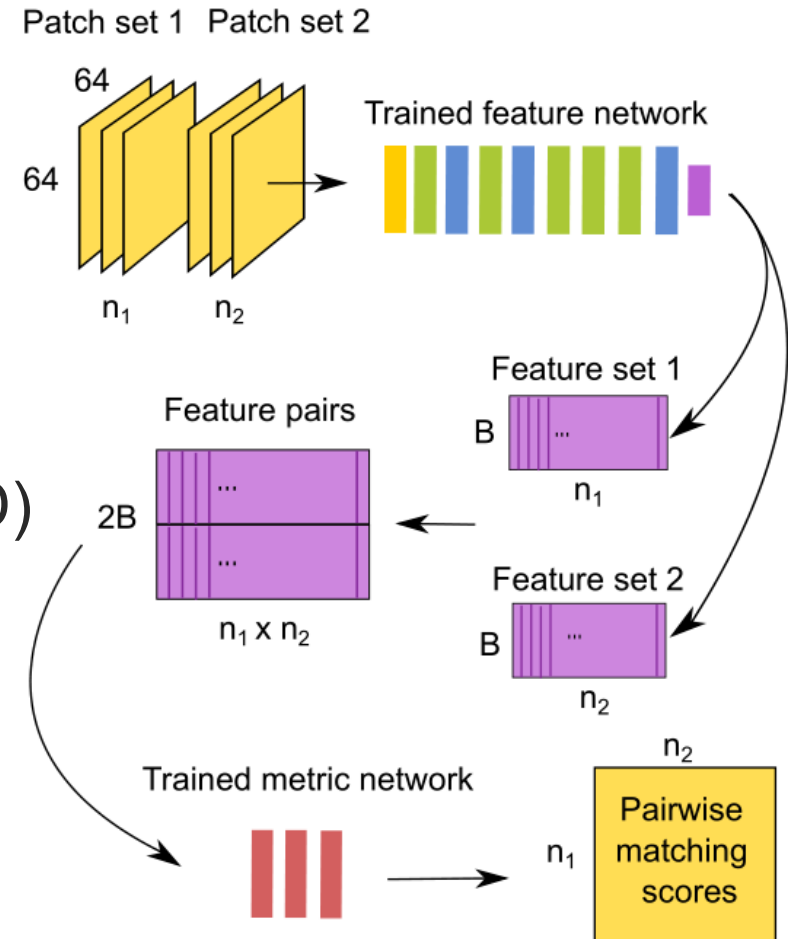
Training MatchNet



- Cross-entropy error

$$E = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

- Stochastic gradient descent (SGD)
- A special reservoir sampler for negative sampling



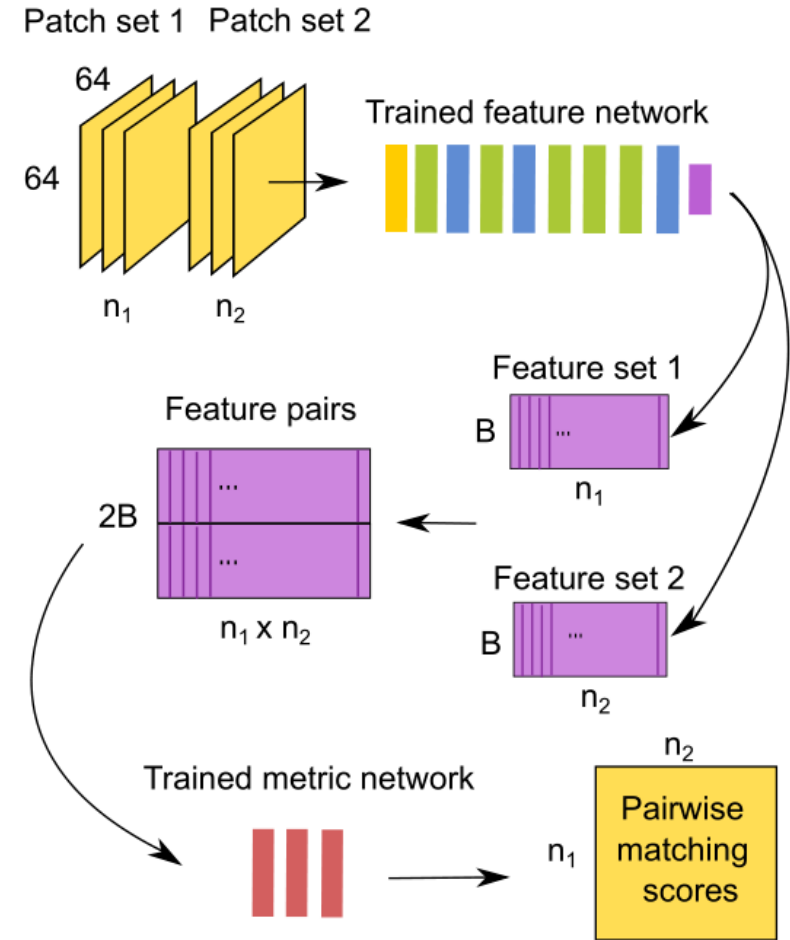


Testing MatchNet



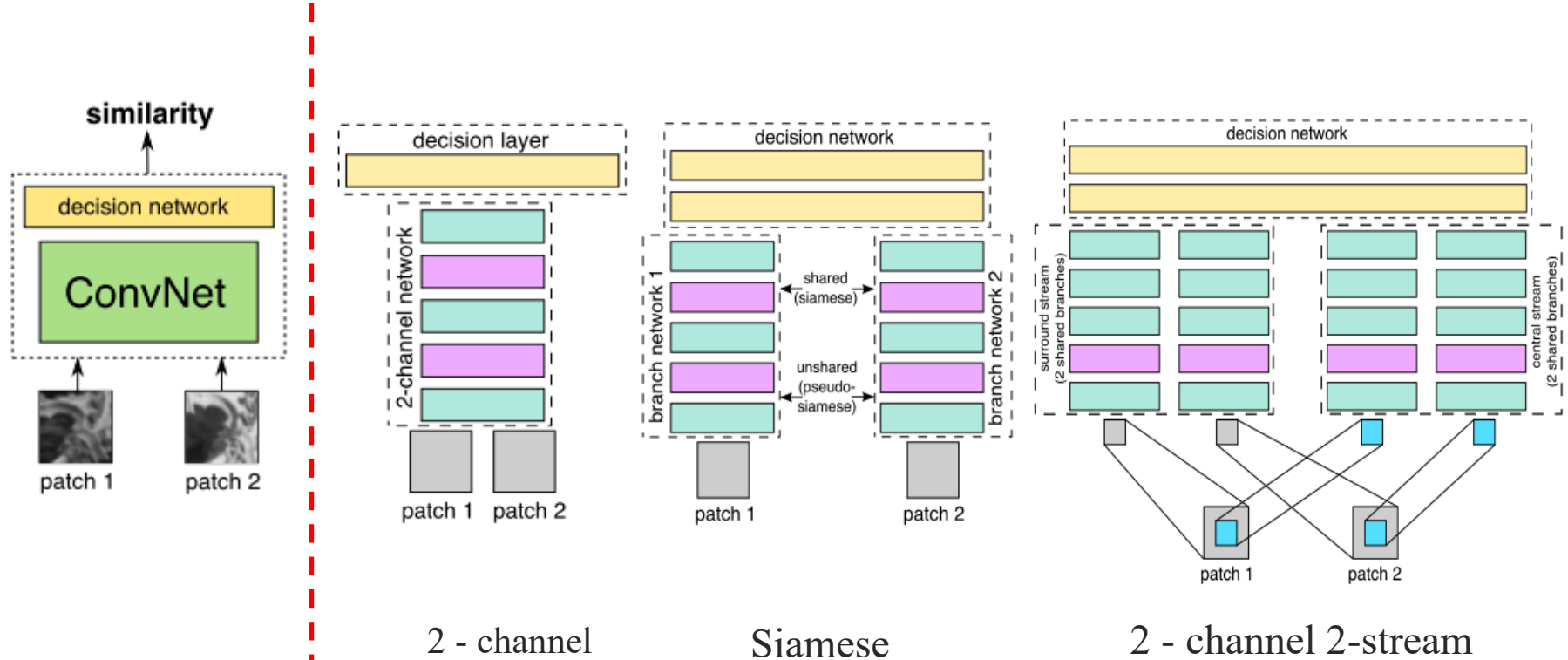
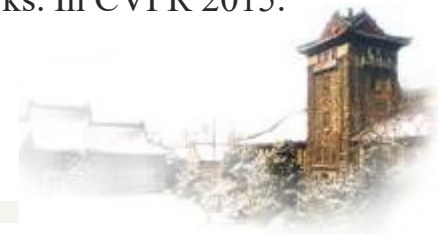
A two-stage prediction pipeline:

1. Generate feature descriptors for all patches.
2. Pair the features and push them through the metric network to get the scores.





DeepCompare





DeepCompare



■ Architecture

○ 2-channel structure

- No direct notion of descriptor in the 2-channel architecture. It simply considers the two patches of an input pair as a 2-channel image, which is directly fed to the first convolutional layer of the network.

○ Central-surround two-stream network

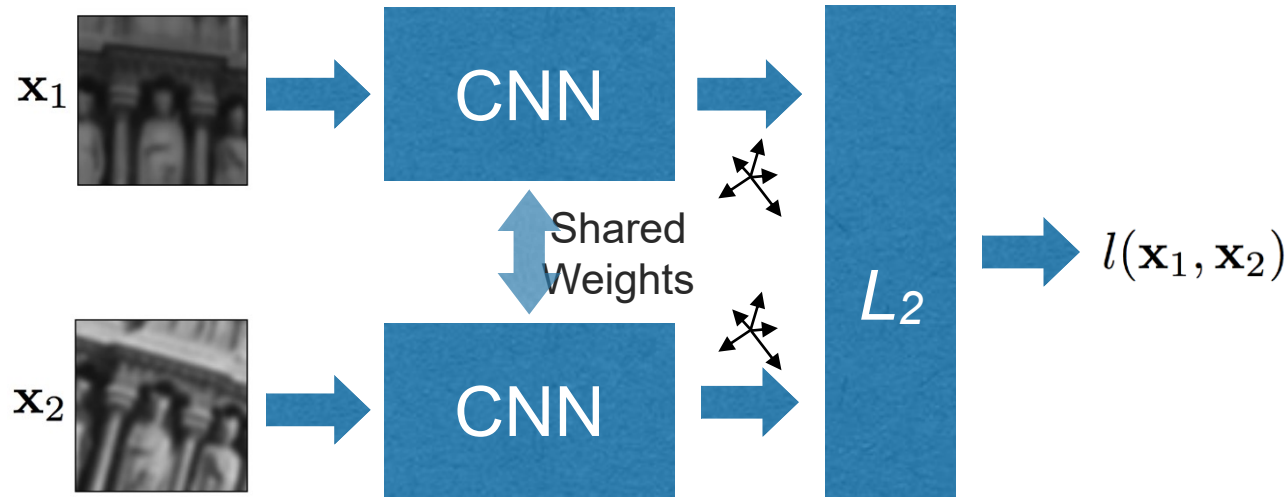
- Consists of two separate streams, central and surround, allowing the network to process at two different resolutions.

■ Drawback

- Pair-wise operation, can not re-use descriptor of each patch



DeepDesc

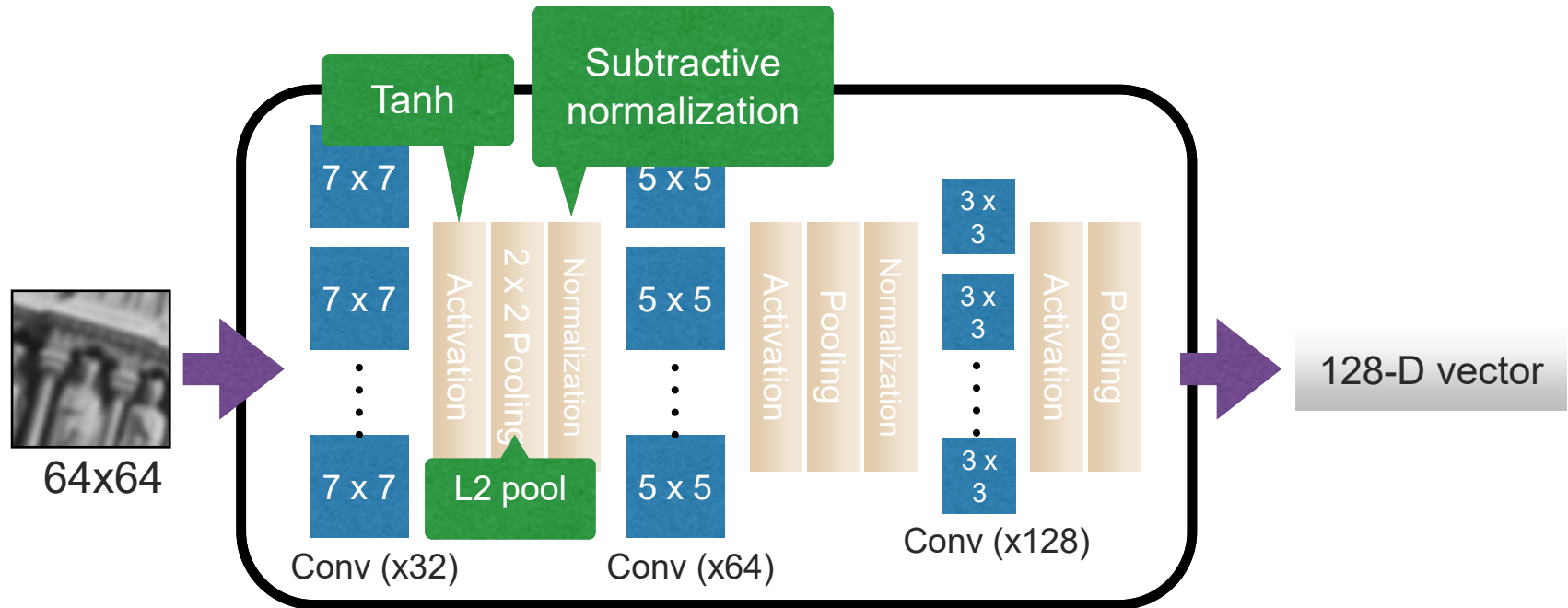


- Use Euclidean distance, direct substitution of SIFT
- loss: minimize pairwise hinge loss

$$l(\mathbf{x}_1, \mathbf{x}_2) = \begin{cases} \|D(\mathbf{x}_1) - D(\mathbf{x}_2)\|_2, & p_1 = p_2 \\ \max(0, C - \|D(\mathbf{x}_1) - D(\mathbf{x}_2)\|_2), & p_1 \neq p_2 \end{cases}$$



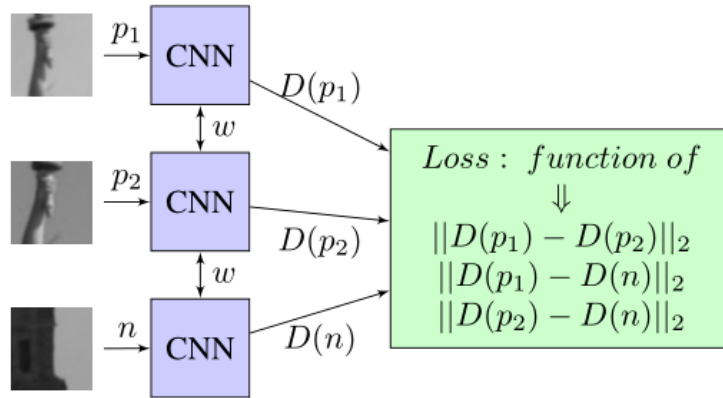
DeepDesc Network Architecture



- Only 3 convolutional layers, simple.
- Use **hard negative mining** to alleviate the problem of imbalanced positive and negative samples, key to good performance.

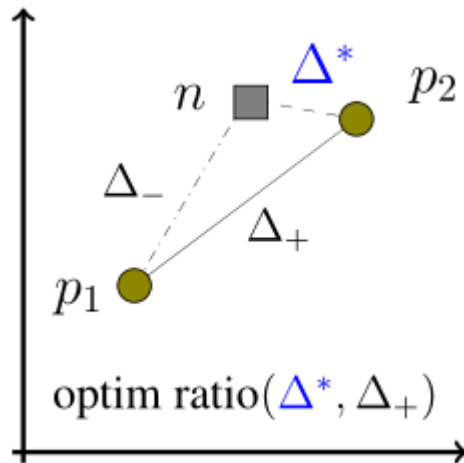


PN-Net, TFeat



CNN Structure

Layer #	Description
1	Spatial Convolution(7,7) $\rightarrow$ 32
2	Tanh
2	MaxPooling(2,2)
3	Spatial Convolution(6,6) $\rightarrow$ 64
4	Tanh
5	Linear $\rightarrow$ {128, 256}
6	Tanh



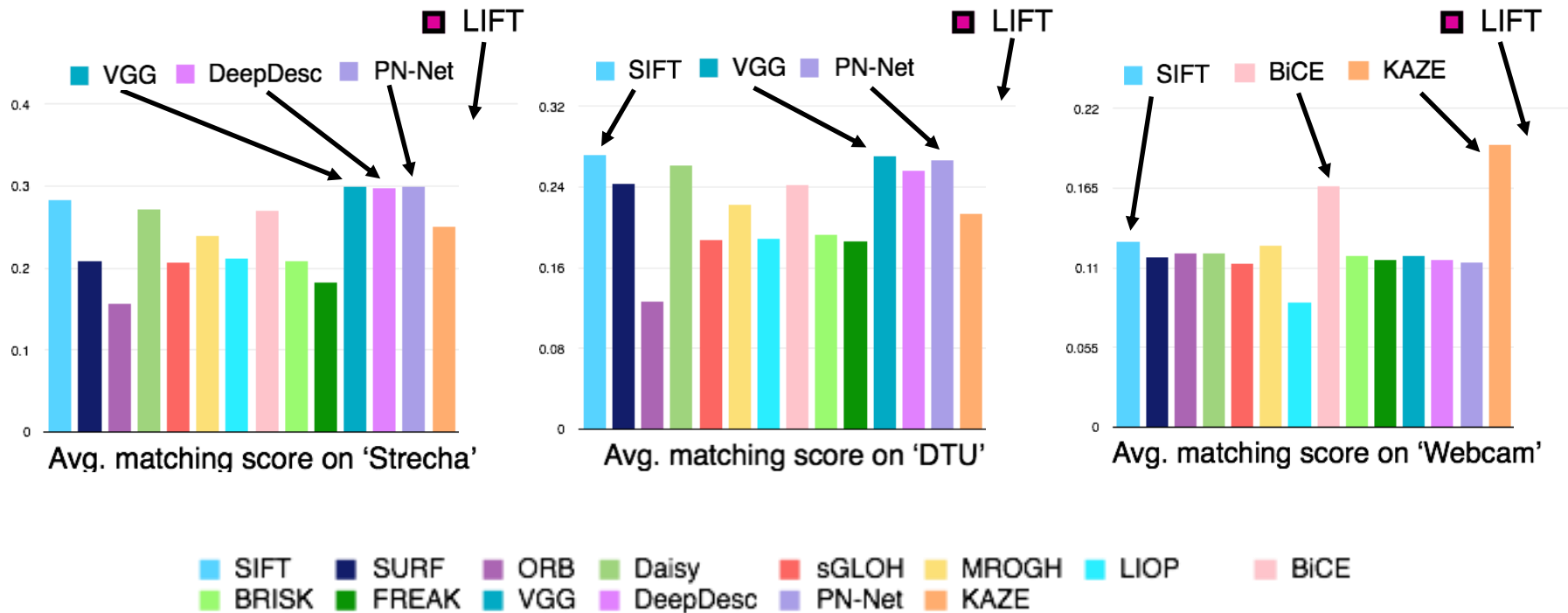
Triplet Network:
Smallest negative distance within the triplet should be larger than the positive distance.



Moreover...



- Descriptors are affected by **keypoints** & **orientations**

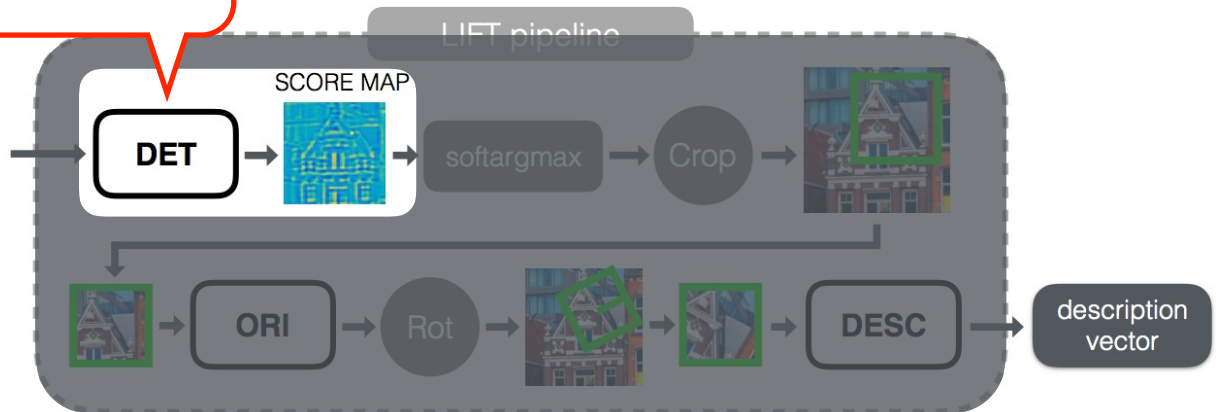
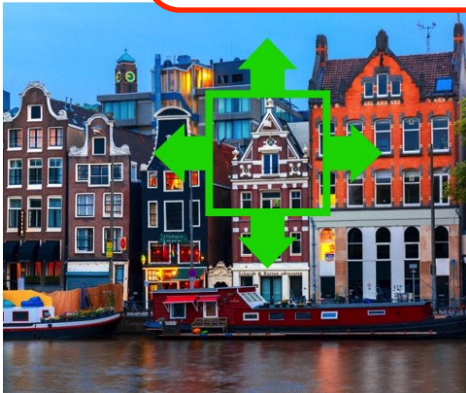




The LIFT Network



Y. Verdie, K.M. Yi, P. Fua, V. Lepetit:
"TILDE: A Temporally Invariant
Learned DETector", CVPR 2015.

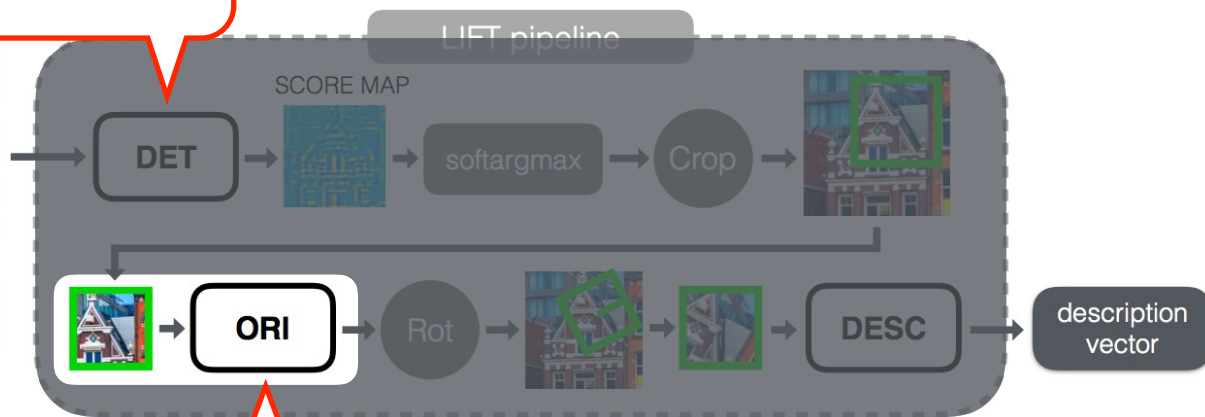
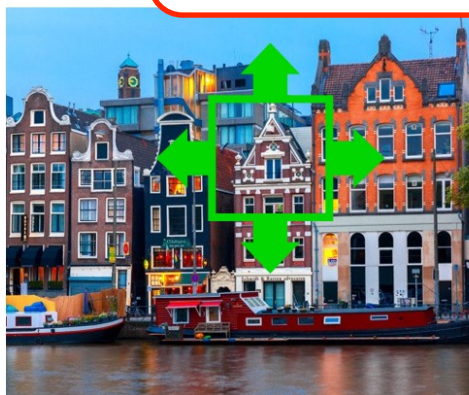




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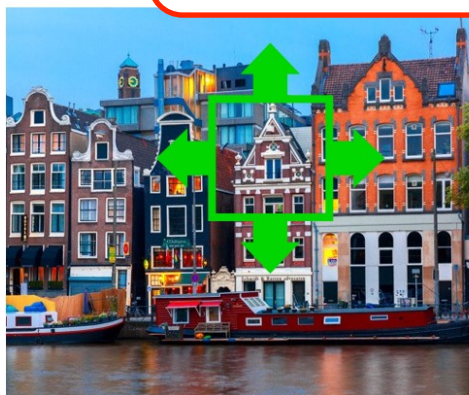
K.M. Yi, Y. Verdie, V. Lepetit, P. Fua :
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E. Simo-Serra, E. Trulls, L. Ferraz, I.
Kokkinos, P. Fua, F. Moreno-Noguer:
"Discriminative Learning of Deep
Convolutional Feature Point
Descriptors", ICCV 2015.

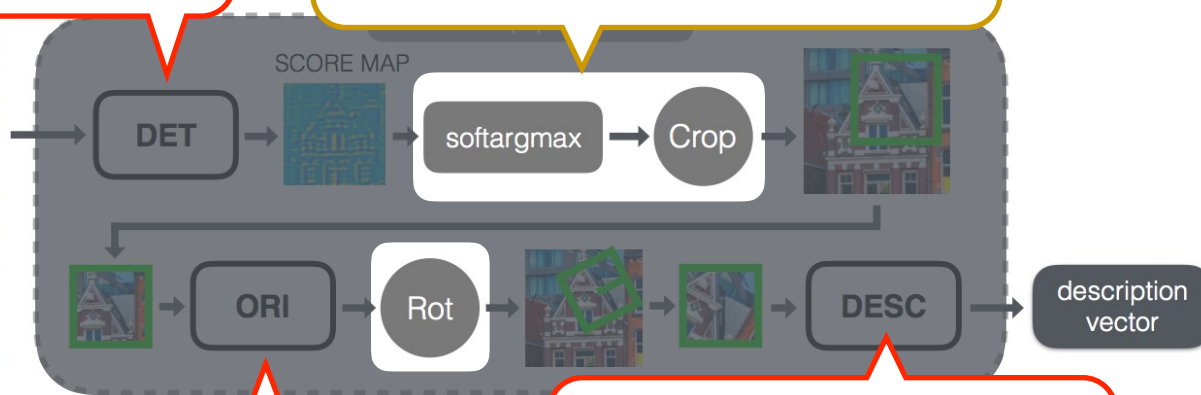
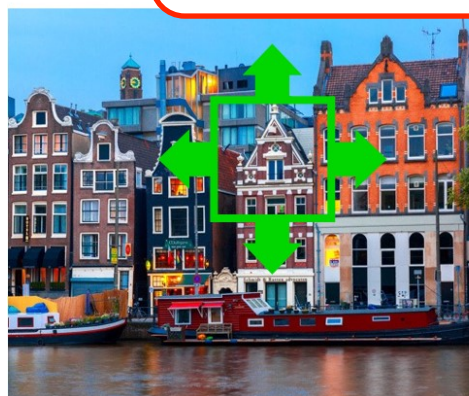


The LIFT Network



Y. Verdie, K.M. Yi, P. Fua, V. Lepetit:
"TILDE: A Temporally Invariant
Learned DEtector", CVPR 2015.

"Glue", to **preserve differentiability**:
• Spatial Transformer Networks, NIPS 2015.
• Soft argmax, Information Retrieval 2009.



K.M. Yi, Y. Verdie, V. Lepetit, P. Fua :
"Learning to Assign Orientations to
Feature Points", CVPR 2016.

E. Simo-Serra, E. Trulls, L. Ferraz, I.
Kokkinos, P. Fua, F. Moreno-Noguer:
"Discriminative Learning of Deep
Convolutional Feature Point
Descriptors", ICCV 2015.



The LIFT Network



Allows
end-to-end
learning

Y. Verdie, K.M. Yi, P. Fua, V. Lepetit:
"TILDE: Temporally Invariant
Learned Invariant Transform",
CVPR 2016.

"Glue", to **preserve differentiability**:

- Spatial Transformer Networks, NIPS 2015.
- Soft argmax, Information Retrieval 2009.

SCORE MAP

softargmax

Crop

Rot

DESC

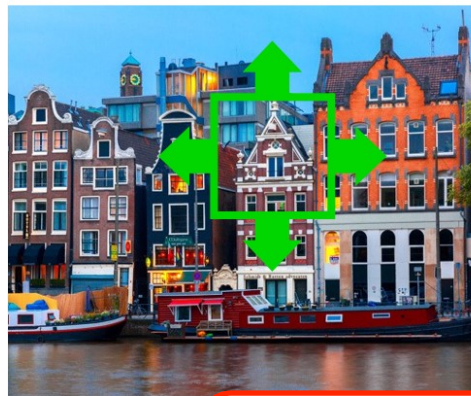
description
vector

K.M. Yi, Y. Verdie, V. Lepetit, P. Fua:
"Learning to Assign Orientations to
Feature Points", CVPR 2016.

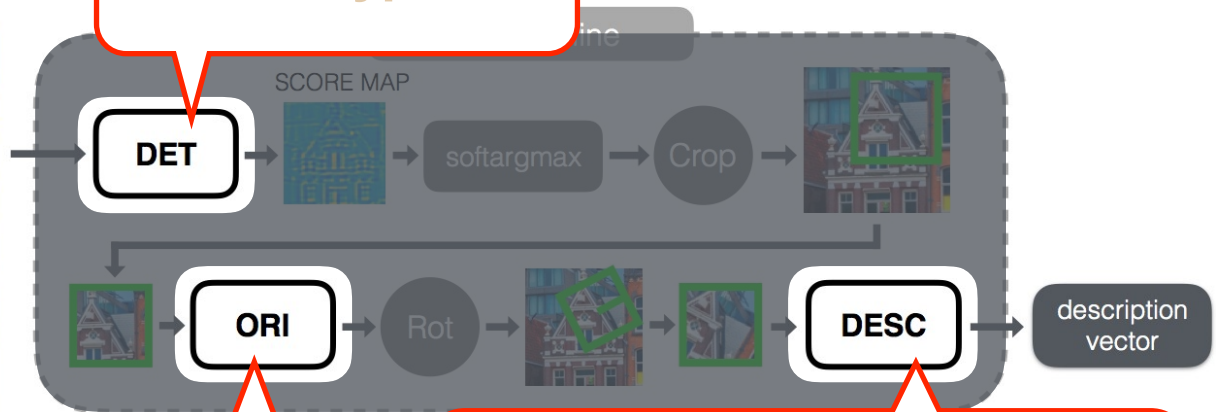
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Kokkinos, P. Fua, F. Moreno-Noguer:
"Discriminative Learning of Deep
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Training requires various patches



Keypoint
VS
Non-Keypoint

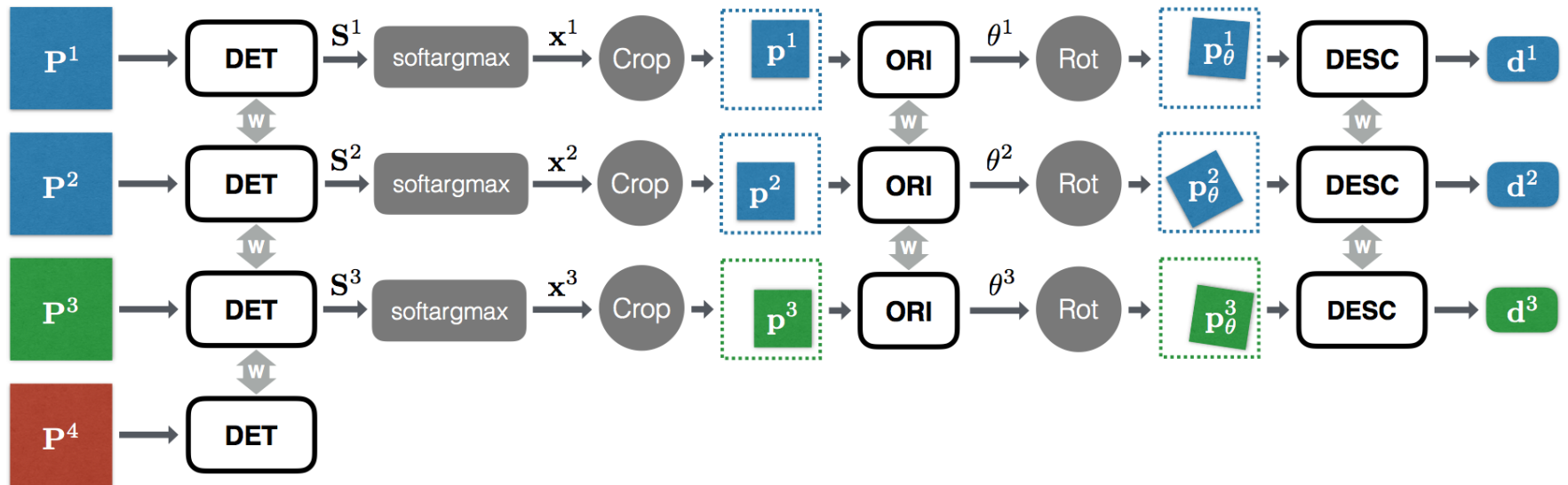


Matching Keypoints

Matching Keypoints
VS
Non-Matching Keypoints



Quadruplet Siamese Network



P_1, P_2 : corresponding keypoints.

P_3 : non-corresponding keypoint.

P_4 : non-keypoint.



A single, global cost function



$$\min_{\{f_\mu, g_\phi, h_\rho\}} \sum_{\{(\mathbf{P}^1, \mathbf{P}^2, \mathbf{P}^3, \mathbf{P}^4)\}} \gamma \mathcal{L}_{class}(\mathbf{P}^1, \mathbf{P}^2, \mathbf{P}^3, \mathbf{P}^4) + \mathcal{L}_{pair}(\mathbf{P}^1, \mathbf{P}^2)$$

detector
orientation
descriptor

$$\mathcal{L}_{class}(\mathbf{P}^1, \mathbf{P}^2, \mathbf{P}^3, \mathbf{P}^4) = \sum_{i=1}^4 \alpha_i \max(0, (1 - \text{softmax}(f_\mu(\mathbf{P}^i)) y_i))^2$$

detector

$$\mathcal{L}_{pair}(\mathbf{P}^1, \mathbf{P}^2) = \| h_\rho(G(\mathbf{P}^1, \text{softargmax}(f_\mu(\mathbf{P}^1)))) - h_\rho(G(\mathbf{P}^2, \text{softargmax}(f_\mu(\mathbf{P}^2)))) \|_2$$

detector
descriptor

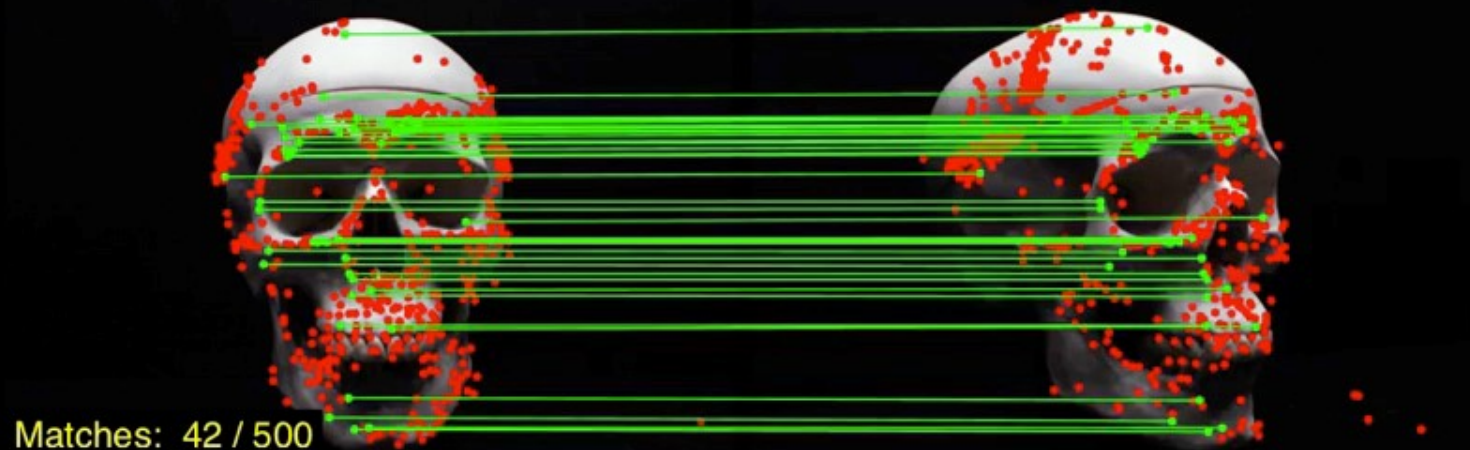
$$G(\mathbf{P}, \mathbf{x}) = \text{Rot}(\mathbf{P}, \mathbf{x}, g_\phi(\text{Crop}(\mathbf{P}, \mathbf{x})))$$

orientation

Matching features on 'DTU', sequence #19.
Correct matches shown with **green** lines.



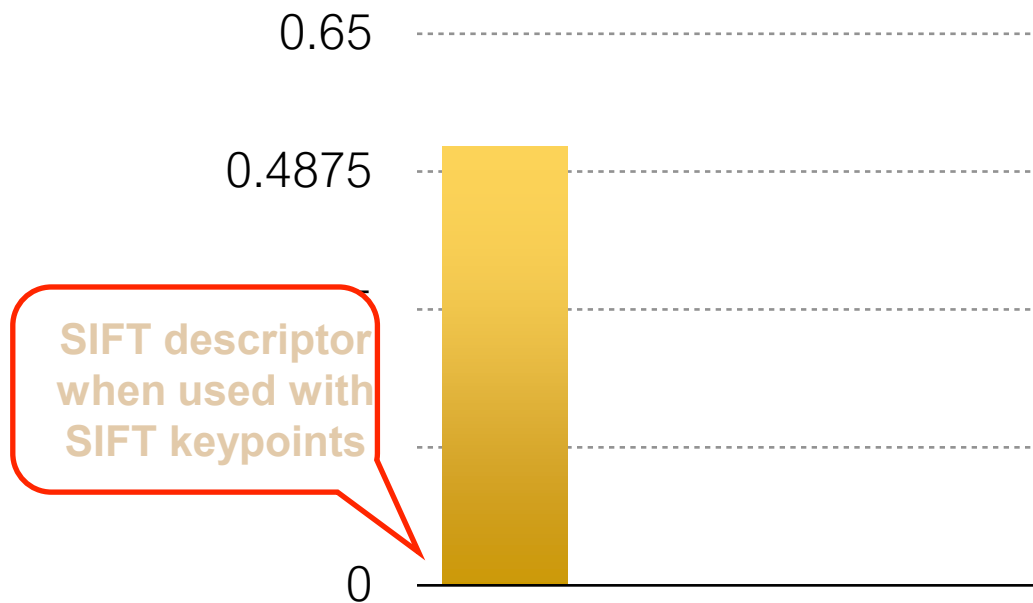
SIFT. Average: **34.1** matches



LIFT (Ours). Average: **98.5** matches



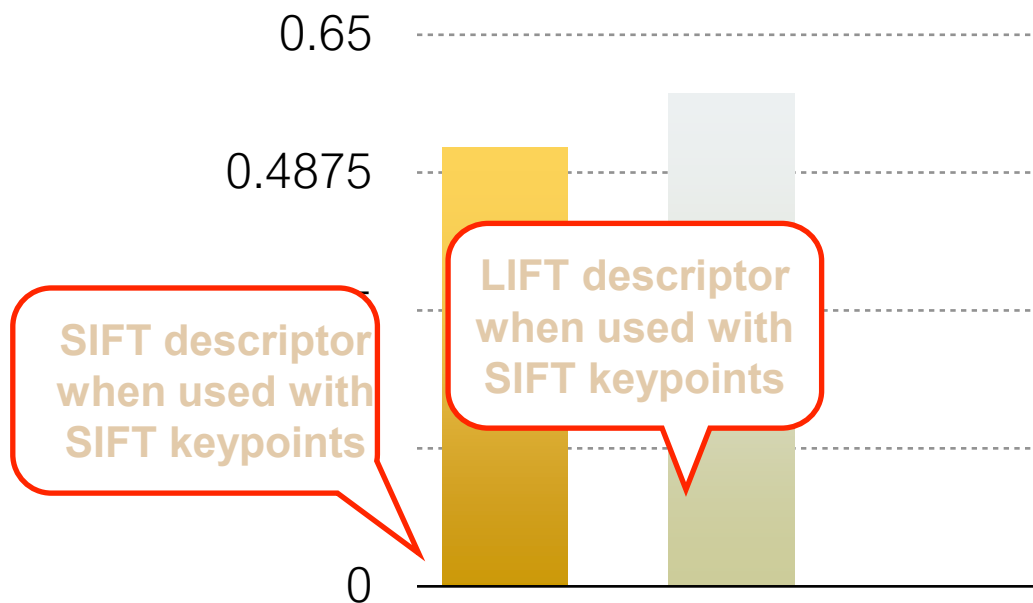
Each component is *meant for* each other



Descriptor performance (NN mAP)



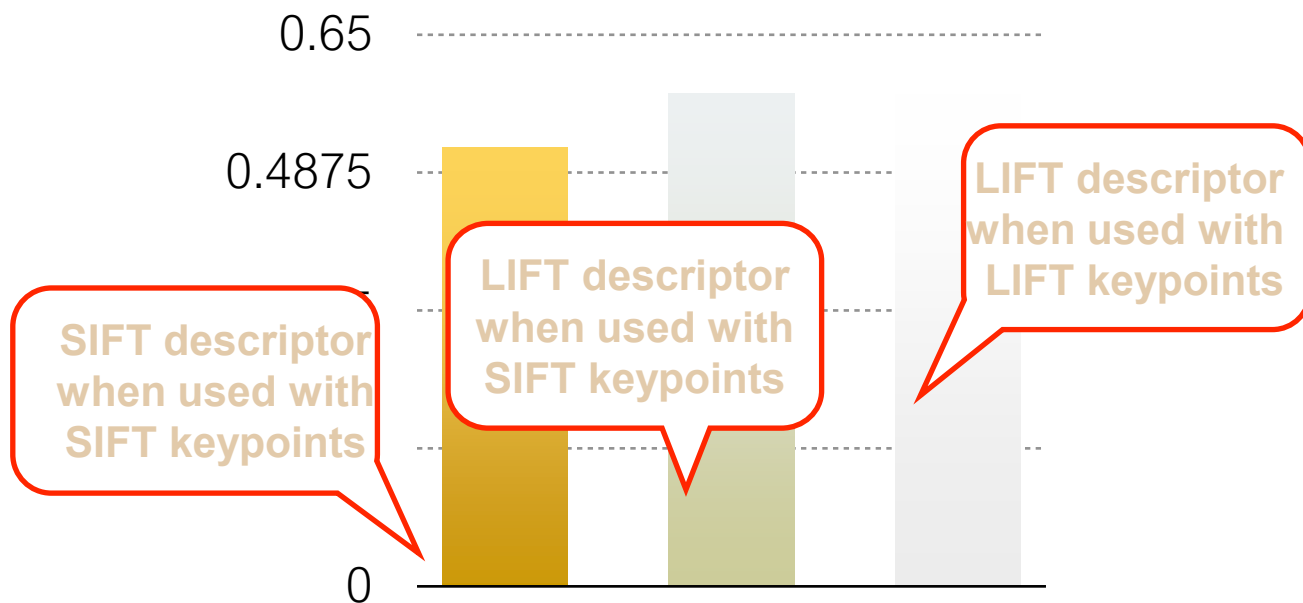
Each component is *meant for* each other



Descriptor performance (NN mAP)



Each component is *meant for* each other



Descriptor performance (NN mAP)



Floating Point Descriptors: A Summarization



- For patch level datasets, **learning based methods generally outperform** hand-crafted ones.
- For image level dataset, **performance gap** between learning based methods and hand-crafted methods **is not significant, except for LIFT**.
- For domain adaptation (e.g. visible to IR), **hand-crafted descriptors are more adaptable**.
- **CNN based methods are dominant** in the learning based methods.
- CNN based methods operating on the **Euclidean space is highly required** for wider application.



Summary: Value of Local Features



- **Advantages**

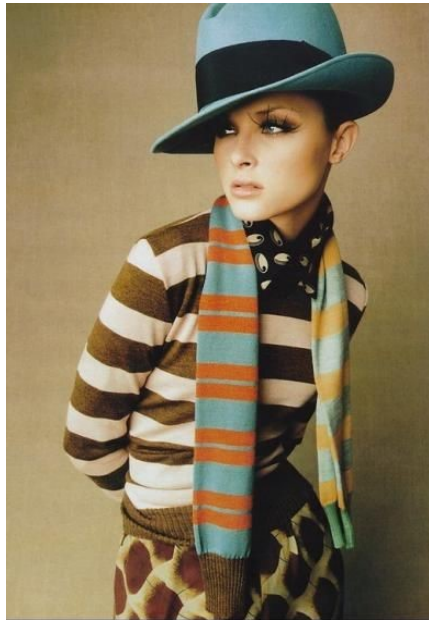
- Critical to find distinctive and repeatable local regions for multi-view matching.
- Complexity reduction via selection of distinctive points.
- Describe images, objects, parts without requiring segmentation; robustness to clutter & occlusion.
- Robustness: similar descriptors in spite of moderate view changes, noise, blur, etc.

Mikolajczyk, K. et al. A comparison of affine region detectors. IJCV, 2005

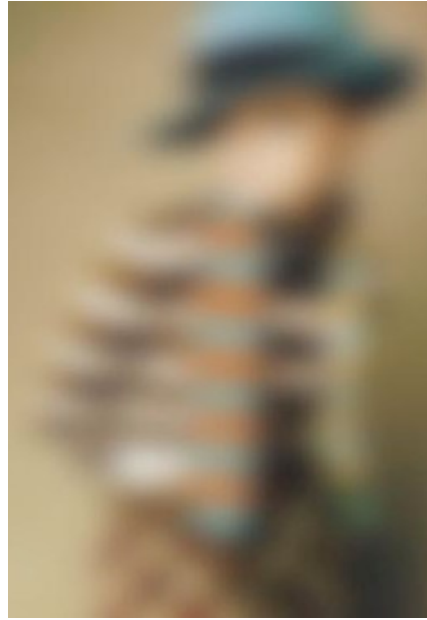
Mikolajczyk, K. et al. A performance evaluation of local descriptors. T-PAMI, 2005.



Discriminative power



Raw pixels



Sampled



Locally orderless



Global histogram

Generalization power



Software



- OpenCV: <http://opencv.org/>
 - SIFT, SURF, BRISK, BRIEF, ORB, FREAK
- VLFeat: <http://www.vlfeat.org/>
 - SIFT, LIOP, Covariant Feature Detectors
- Authors' pages, Github, etc.
- Supplementary containing implementation information
 - Learned Orientations: <https://infoscience.epfl.ch/record/217982/files/0141-sup.pdf>
 - LIFT: https://documents.epfl.ch/groups/c/cv/cvlab-unit/www/data/keypoints/lift/paper_1377_supplementary.pdf



Image Matching Benchmark and Challenge

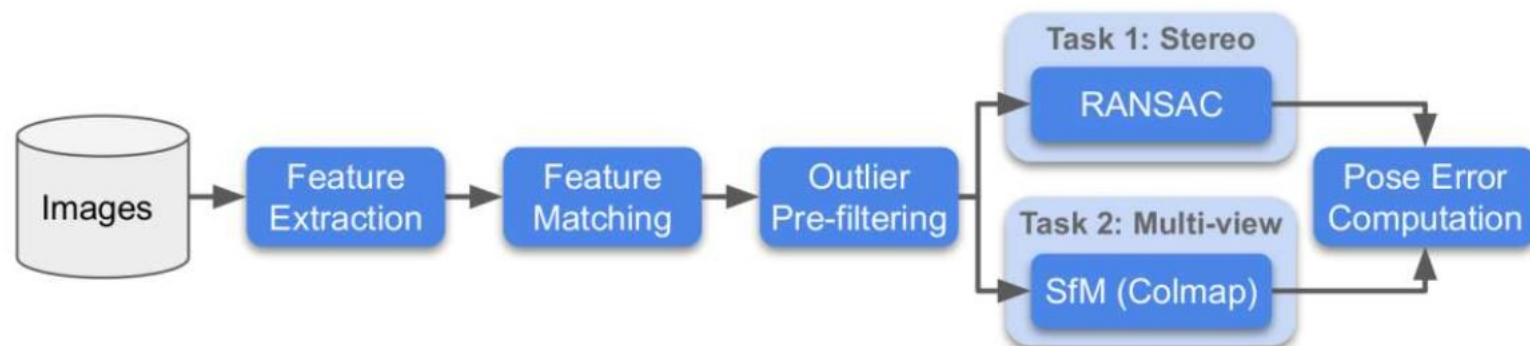
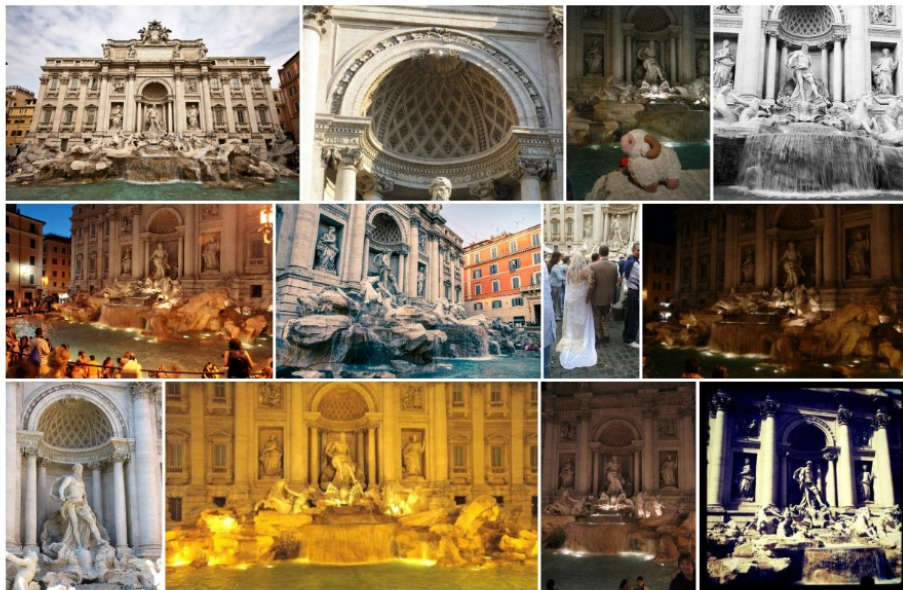




Image Matching Benchmark and Challenge

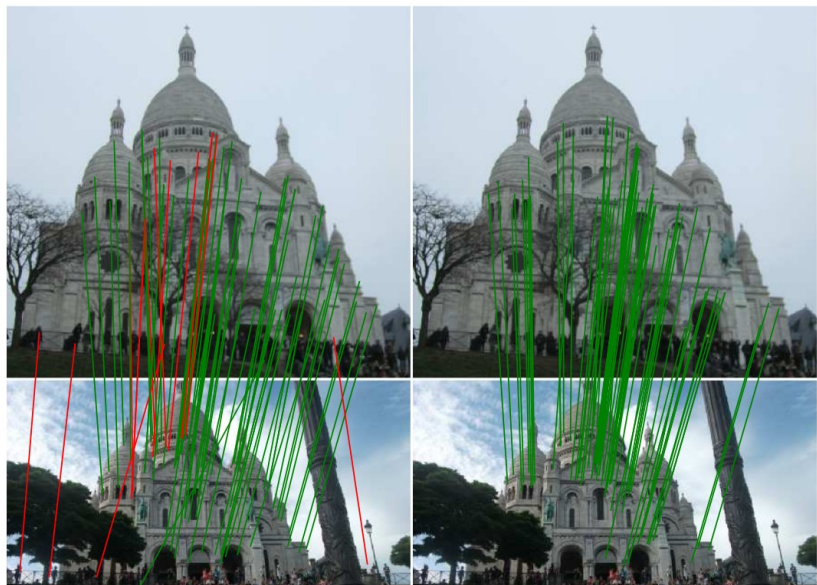


Figure 1. Every paper claims to outperform the state of the art. Is this possible, or an artifact of insufficient validation? On the left, we show stereo matches obtained with **D2-Net** (2019) [33], a state-of-the-art local feature, using OpenCV RANSAC with its default settings. On the right, we show **SIFT** (1999) [48] with a carefully tuned MAGSAC [29] – notice how the latter performs much better. We fill this gap with a new, modular benchmark for sparse image matching, with dozens of built-in methods.

Method	PyRANSAC			DEGENSAC		MAGSAC		Rank
	NF	NI [†]	mAP(10°) [†]	NI [†]	mAP(10°) [†]	NI [†]	mAP(10°) [†]	
CV-SIFT	7879.0	153.9	.4160	222.7	.4608	270.6	.4586	13
VL-SIFT	7901.0	166.2	.4137	241.2	.4643	301.0	.4638	12
VL-Hessian-SIFT	8000.0	186.5	.3915	264.0	.4489	318.0	.4394	15
VL-DoGAff-SIFT	7910.9	218.6	.4049	229.7	.4653	291.9	.4624	11
VL-HesAffNet-SIFT	8000.0	190.8	.4081	271.6	.4659	328.5	.4581	10
CV- $\sqrt{\text{SIFT}}$	7884.0	176.3	.4348	257.4	.4921	317.4	.4891	6
SURF	7749.0	113.0	.2326	117.8	.2452	136.1	.2481	19
AKAZE	7879.8	184.3	.2738	232.7	.3142	284.4	.3054	17
ORB	7128.2	113.1	.1381	136.6	.1723	163.2	.1632	22
DoG-HardNet	7884.1	229.9	.4668	342.2	.5286	404.0	.5147	1
DoG-HardNetAmos+	7884.1	213.6	.4511	316.9	.5125	373.5	.5011	3
L2Net	7884.8	190.2	.4478	280.9	.4971	329.1	.4884	5
Key.Net-HardNet	7998.1	353.3	.3990	375.2	.4700	636.5	.4529	9
Geodesc	7884.3	179.7	.4183	264.0	.4787	340.3	.4753	8
ContextDesc	4811.1	248.8	.4283	261.4	.4856	356.1	.4662	7
SOSNet	7884.3	215.1	.4595	319.8	.5233	418.5	.5177	2
LogPolarDesc	7884.3	243.5	.4495	366.0	.5080	461.0	.5001	4
SuperPoint (2k)	1178.9	88.1	.2359	84.7	.2669	113.2	.2620	18
LF-Net (2k)	2024.8	95.1	.1945	100.8	.2253	134.2	.2164	20
D2-Net (SS)	5540.7	273.5	.1432	241.4	.1639	428.0	.1560	23
D2-Net (MS)	6806.3	193.8	.1690	322.8	.1836	505.1	.1731	21

Table 1. **Stereo – Test set.** We report: **(NF)** Number of Features; **(NI)** Number of Inliers produced by RANSAC; and **mAP(10°)**. Top three methods by mAP marked in **red**, **green** and **blue**.